6.3 Function Fields and the Riemann-Roch Theorem

We develop the theory of algebraic function fields K/k, which by (6.3.4) are the same thing as complete regular algebraic curves over k which are geometrically integral. Our goal is the Riemann-Roch theorem for such curves over a perfect base field k, the algebraic counterpart of the classical theorem for compact Riemann surfaces. Via the correspondence it suffices to prove it for function fields, where it was proved by Weil and Chevalley [Che51] using adèles (répartitions). That proof needs the least machinery, but in it the genus g and the index of speciality $i(D)$ appear only as extremal or dimension-count invariants, with no direct geometric meaning. As in the classical case, the remaining work is to relate them to rational differentials: for a non-zero rational differential ω with divisor (ω) we show $i(D) = \ell((\omega) - D)$ and $g = \ell((\omega))$. This rests on the Residue Theorem $\sum_P \operatorname{Res}_P(\omega) = 0$ (6.3.65), which we need over a general perfect field and not only for $k = \overline{k}$. The closed points of a curve over k have residue fields $k(P)$ of arbitrary finite degree over k. At such a place we expand in $k(P)((X))$ and set $\operatorname{Res}_P := \operatorname{Tr}_{k(P)/k} \circ \operatorname{Res}_P^{k(P)}$ (6.3.37); equivalently Res_P is the trace of the residue at a conjugate rational place over $\overline{k}$ (6.3.64). A rational differential ω defines a Weil differential by $\alpha \mapsto \sum_P \operatorname{Res}_P(\alpha_P \omega)$, whose vanishing on K is exactly the Residue Theorem. We deduce it from the algebraically closed case (6.3.46) by base change to $\overline{k}$: the places of K correspond to Galois orbits of places of $K_{\overline{k}}$, and the residue at P is the sum of the residues over its orbit.

We first fix the basic notions for function fields (an algebraic function field, its field of constants and its places) and then state the correspondence with curves precisely.

Proposition 6.3.1
Let K/k be a field extension. Then the following are equivalent

a) K/k is finitely generated with transcendence degree 1.
b) There exists a transcendental element $x \in K/k$ such that $K/k(x)$ is finite.

*In this case we say that K/k is an **algebraic function field**. We typically assume that k is perfect (either a number field or a finite field).*

Proof. *a)* $\Longrightarrow$ *b)* By (3.18.139) a transcendence base consists of a single element x. By assumption $K/k(x)$ is algebraic and finitely generated, hence also finite (3.18.44).

b) $\Longrightarrow$ *a)* Evidently $\{x\}$ is a transcendence base. By assumption $K/k(x)$ is finite and therefore certainly finitely generated, whence so is K/k. ∎

Definition 6.3.2 (Algebraic Function Field)
*Let K/k be an algebraic function field. Denote by $\tilde{k}$ the **field of constants**, the relative algebraic closure of k in K:*

$$\tilde{k} := \{x \in K \mid x \text{ algebraic over } k\}$$

*(a subfield of K by (3.18.45)). By (3.18.46), $K/\tilde{k}$ is relatively algebraically closed. We say that k is the **full constant field** of K/k if $k = \tilde{k}$.*

Definition 6.3.3 (Place of an Algebraic Function Field)
*Let K/k be an algebraic function field. By a **place** of K/k we refer to a valuation ring $\mathcal{O}_P$ such that $k \subsetneq \mathcal{O}_P \subsetneq K$. By (3.24.2) this is a local ring with unique maximal ideal $\mathfrak{m}_P$.*

From (3.3.13) we verify immediately that $k \cap \mathfrak{m}_P = (0)$.

*For every $x \in K$ we say that x is **regular** at a place P if $x \in \mathcal{O}_P$.*

Theorem 6.3.4 (Curves and Function Fields)
Let k be a field. The assignment $X \mapsto k(X)$ is an anti-equivalence between

- *complete, regular, geometrically integral curves X over k, with non-constant morphisms, and*
- *algebraic function fields K/k in which k is relatively separably closed, with k-embeddings.*

Here X is geometrically integral (6.2.8) if and only if k is relatively separably closed in $k(X)$ (6.2.9). When k is perfect this says that k is the full constant field of $k(X)$, that is $\tilde{k} = k$.

Writing X_0 for the Kolmogorov quotient of the points of X, under this correspondence

a) there is a bijection from X_0 to the set of places of $k(X)/k$, $x_0 \mapsto P$, such that $\mathcal{O}_{X,x} = \mathcal{O}_P$ and $k(x) = k(\mathfrak{m}_P)$;
b) a non-constant morphism $\phi : X \to Y$ corresponds to the embedding $\phi_\star : k(Y) \hookrightarrow k(X)$, with $\deg(\phi) = [k(X) : \phi_\star(k(Y))]$, and $\phi_0(x_0) = y_0$ if and only if $\phi_\star(\mathcal{O}_{Y,y}) \subset \mathcal{O}_{X,x}$;
c) a quasi-inverse sends K/k to a curve X_K with a bijection from $(X_K)_0$ to the set of places of K/k as in a).

© David Rufino CC BY-NC-ND 4.0

The proof is given at the end of the next subsection, once the places of an algebraic function field and their relation to primes of integral closures have been studied.

6.3.1 Relation to the Literature

These treatments fall into two methodological groups:

- Weil's répartition/adele method, in the tradition initiated by Chevalley [Che51]: used by Chevalley himself, by Stichtenoth [Sti09], by Lang [Lan82] (all three over a general constant field), and, at its computational core, even by Serre [Ser59] – whose own bibliographic note attributes the proof to Weil, though he additionally erects the formalism of sheaf cohomology around it, identifying $H^1(X, \mathcal{O}_X)$ with the classes of repartitions. Lang's book, unlike the others in this group, has a different overall aim: a bridge to the classical complex-analytic theory of abelian and theta functions, which the algebraic function-field theory here does not pursue.
- the classical algebraic-geometric (scheme-free, ringed-space) approach shared by Hartshorne [Har13] and Bump [Bum98], the latter specialised to curves with a dedicated treatment of differentials, residues and ramification.

I have interpolated between these, guided by four principles.

Dedekind domains over bare valuation theory. I use Dedekind domain theory for the key results on function fields and the Weil form of Riemann-Roch, rather than the purely valuation-theoretic development of [Che51] and [Sti09]. In the same spirit, finite extensions of $k((t))$ (3.32.35) are analysed directly via the integral closure in a Dedekind domain and a direct power-series computation, rather than via the P-adic completion machinery both of those references build on; this also yields the result in every characteristic, where [Che51]'s analogous Puiseux-expansion treatment (Ch. II, §6) is restricted to the tame case, as noted in the remark opening the Extensions of Power Series Rings subsection. We also use the theory of affine algebras to simplify the proof of some further technical results.

A perfect constant field, reduced to the algebraically closed case only once. Riemann-Roch is proved over a general constant field, matching [Sti09], but going beyond [Ser59], [Har13], [Bum98] and [Lan82], whose treatments of Riemann-Roch all assume an algebraically closed constant field throughout. The Residue Theorem, on which the identification of the genus and the index of speciality with rational differentials rests, is likewise stated in the literature only over an algebraically closed field ([Ser59] (Ch. II), [Lan82] (Ch. I, §5), [Sti09] and [Bum98]). Our proof in that case (6.3.46) follows [Ser59] (Ch. II, nos. 12–13) and [Lan82] (Ch. I, §§4–5): partial fractions on the rational function field, the trace formula for residues to reduce the general case to it, and a local trace formula for power series fields proved by a universal identity over $\mathbb{Z}$. The local theory rests on the structure of finite extensions of $k((t))$ from the Extensions of Power Series Rings subsection (3.32.35), (3.32.36) and the Weierstrass division lemma (3.32.31), as [Lan82] (Ch. I, §4, Prop. 4.3) does. We need the Residue Theorem over a perfect field (6.3.65), and deduce it from the algebraically closed case by relating the residue symbol over k to its base change to $\overline{k}$: the residue at a place P is the trace of the residue at a conjugate place of $K_{\overline{k}}$ above it (6.3.64), so the sum of the residues at the places of K is the sum of the residues at all the places of $K_{\overline{k}}$ (6.3.63).

The perfect case can also be assembled using only results of [Sti09]. There the residue is defined only at places of degree one (Def. 4.2.10), and the Residue Theorem (Cor. 4.3.3) is stated only over an algebraically closed field, where every place has degree one. But Thm. 4.3.2 identifies differentials with Weil differentials over any perfect constant field K, and for a place P of degree f, with the f places $\overline{P}_1, \dots, \overline{P}_f$ of the constant field extension $F\overline{K}$ above it (Cor. 3.6.5), Rem. 4.3.7(d) gives the formula (4.38)

$$\omega_P(u) = \sum_{i=1}^{f} \operatorname{res}_{\overline{P}_i}(u\omega),$$

that is, the local component of a differential at P is the sum of the residues at the conjugate rational places. This is exactly our residue symbol at P (6.3.63). The vanishing of the sum is then Prop. 1.7.2, $\sum_P \omega_P(1) = \omega(1) = 0$, valid over any constant field since a Weil differential vanishes on the principal adele 1. So over a perfect field the Residue Theorem is literally $\sum_P \omega_P(1) = 0$ once residues are read as local components; its algebraically closed statement is the special case where every place is rational. The compatibility of local components with constant field extension used there is Prop. 4.3.6, (4.33), which rests on the cotrace formula (4.21) – the cotrace of a Weil differential ω of F in a finite separable extension F' being the Weil differential ω' of F' characterised by $\omega_P\big(\operatorname{Tr}_{F'/F}(y)\big) = \sum_{P'|P} \omega'_{P'}(y)$ for $y \in F'$ – and the construction of the Weil differential $\delta(x)$ as a cotrace; our approach replaces it by the direct comparison of Laurent expansions (6.3.62).

© David Rufino CC BY-NC-ND 4.0

Concrete constructions over general machinery, wherever one suffices. Like [Che51], [Sti09] and [Lan82], I work with Chevalley and Weil's répartition/adele-style methods directly, and connect this to cohomology only after the fact, rather than building up the general machinery of coherent sheaf cohomology as [Ser59] does around the same underlying répartition computation. Residues are defined directly from the Laurent expansion at a place, via the uniformiser derivation, without ever constructing the module of Kähler differentials $\Omega_{K/k}$, in contrast to [Ser59] and [Sti09], which both identify their (Weil-differential-based) residues with genuine differentials in this sense; at a non-rational place the residue is the trace of the residue in $k(P)((X))$, rather than a local component of a Weil differential defined through cotraces as in [Sti09]. As in [Lan82], the local analysis uses only explicit Laurent series fields and the structure of their finite extensions (3.32.35), rather than the abstract P-adic completion on which [Ser59] and [Sti09] depend, and the rational function field is handled by a short, self-contained proof by partial fractions rather than as a special case of the general machinery. The Trace Formula for Residues (6.3.54) likewise rests on this Laurent-series finite-extension lemma, and not on the general Complete Discrete Valuation Rings section developed alongside it, of which the lemma is the special case $\mathcal{O}_K \cong k[[t]]$.

Characteristic p handled uniformly, via universal $\mathbb{Z}$-polynomial identities. Invariance of the residue of a differential under change of uniformiser – notoriously tricky in characteristic p, where the naive derivative-of-a-power argument breaks down whenever p divides the relevant exponent – is settled once, universally, over $\mathbb{Z}[a_1, a_1^{-1}, a_2, a_3, \ldots]$ and specialised (6.3.38), the same device used by [Ser59] and [Sti09], rather than the Cartier-operator route of [Bum98]; unlike [Ser59] and [Sti09], this is done over a general constant field rather than an algebraically closed one. The same device is reused for the Trace Formula for Residues, which holds without any tameness hypothesis (6.3.52) by the identical method [Ser59] uses (there attributed to Hasse) for the analogous local identity $\mathrm{Res}_t(f\,du) = \mathrm{Res}_u(\mathrm{Tr}(f)\,du)$ underlying its own residue formula; it is exactly this universal-identity device that lets the wildly ramified case go through in both results.

Two further points of departure concern scope and connections to the classical picture, rather than proof technique: the dictionary between the function field and the regular algebraic curve – established even over a non-algebraically closed field, generalising the algebraically-closed-field statement of [Har13] – is established earlier, in (6.2.120) of the Algebraic Geometry chapter's treatment of regular algebraic curves; and, in the classical case of a compact Riemann surface (6.3.31), Weil differentials are shown to coincide canonically with classical meromorphic 1-forms, reconnecting the algebraic formalism with the analytic theory it generalises.

6.3.2 Algebraic Function Fields

We develop the basic theory of **algebraic function fields** of one variable, the field-theoretic counterpart to a regular algebraic curve.

Proposition 6.3.5
Let K/k be a function field and $k \subsetneq \mathcal{O}_P \subsetneq K$ a valuation ring. Then $\tilde{k} \subset \mathcal{O}_P$ and $\tilde{k} \cap \mathfrak{m}_P = (0)$

Proof. By definition $\tilde{k}$ is the integral closure of k in K, so $\tilde{k} \subset \mathcal{O}_P$ by (3.24.14). Suppose $0 \neq x \in \tilde{k}$ then x is invertible in $\mathcal{O}_P$ and therefore x is not contained in any proper ideal, and in particular $x \notin \mathfrak{m}_P$. ∎

Proposition 6.3.6 (Residue Degree of a Place)
Let K/k be an algebraic function field and $f \in K \setminus \tilde{k}$. Then

a) $K/k(f)$ is finite.
b) If $\mathcal{O}_P$ is a place with $f \in \mathcal{O}_P$, then $\mathfrak{p} := \mathfrak{m}_P \cap k[f]$ is a maximal ideal of $k[f]$, the field $k(\mathfrak{p}) := k[f]/\mathfrak{p}$ is finite over k, and

$$[k(\mathfrak{m}_P) : k(\mathfrak{p})] \leq [K : k(f)]$$

c) Consequently the residue field $k(\mathfrak{m}_P)$ of every place P of K/k, whether or not it contains f, is a finite extension of k.

Proof. a) As f is not algebraic over k and $\mathrm{trdeg}(K/k) = 1$ (6.3.1), f is a transcendence base (3.18.139), so $K/k(f)$ is algebraic and finitely generated, hence finite (3.18.44).

b) The ring $k[f]$ is a polynomial ring, hence a PID (3.15.4), and $\mathfrak{p}$ is a prime ideal of it contained in $\mathfrak{m}_P$. As $\mathcal{O}_P \neq K$ is not a field, $\mathfrak{m}_P \neq 0$, and every element of $\mathcal{O}_P \subset K$ is algebraic over $k(f)$ by a), so $\mathfrak{p} \neq 0$ (3.23.17). So $\mathfrak{p}$ is a non-zero prime ideal of a PID, hence maximal, and $\mathfrak{p} = (g(f))$ for an irreducible polynomial g. Thus $k(\mathfrak{p}) \cong k[X]/(g)$ is finite over k (3.18.40). Further $k[f]_{\mathfrak{p}}$ is a discrete valuation ring (3.30.4), in particular a PID, and $\mathcal{O}_P$ is an overring of $k[f]$ with quotient field K containing the proper ideal $\mathfrak{m}_P \supsetneq \mathfrak{p}$. So (3.31.18) applied to $k[f] \subset \mathcal{O}_P$ gives the inequality.

c) As $\mathcal{O}_P \neq K$ is not a field, there is $0 \neq x \in \mathfrak{m}_P$. Then $x \notin \tilde{k}$ by (6.3.5), so b) applies to x, and $[k(\mathfrak{m}_P) : k] = [k(\mathfrak{m}_P) : k(\mathfrak{p})][k(\mathfrak{p}) : k]$ is finite. ∎

© David Rufino CC BY-NC-ND 4.0

In order to use the theory of Dedekind domains and affine algebras we establish a connection between valuation rings and a certain "affine" subring.

Proposition 6.3.7
Let K/k be an algebraic function field and $f \in K \setminus \tilde{k}$ transcendental. Then the integral closure $\overline{k[f]}$ in K is a Dedekind domain with quotient field K and a finite $k[f]$-module. There is a correspondence

$$\left\{\mathfrak{B} \lhd \overline{k[f]} \;\middle|\; \mathfrak{B} \text{ non-zero prime}\right\} \longleftrightarrow \{\mathcal{O}_P \subsetneq K \mid f \in \mathcal{O}_P\}$$
$$\mathfrak{B} \longrightarrow \overline{k[f]}_{\mathfrak{B}}$$
$$\mathfrak{m}_P \cap \overline{k[f]} \longleftarrow \mathcal{O}_P$$

In particular every such $\mathcal{O}_P$ is a discrete valuation ring, and $k(\mathfrak{m}_P) = k(\mathfrak{B})$. Moreover every place P of K/k (whether or not $f \in \mathcal{O}_P$) is a discrete valuation ring, so we may associate to it a discrete valuation

$$v_P : K \longrightarrow \mathbb{Z} \cup \{\infty\}$$

which is zero on k and $v_P(f) < 0 \iff f \notin \mathcal{O}_P \iff f^{-1} \in \mathfrak{m}_P$.

Proof. As $f \in K \setminus \tilde{k}$ is transcendental over k, $k[f]$ is a polynomial ring and so a PID (3.15.4), hence an affine Dedekind domain (3.31.2), which has Krull dimension 1 (3.31.1). Also $K/k(f)$ is finite (6.3.6).a). By (3.34.61) $B := \overline{k[f]}^K$ is therefore a Dedekind domain and finite $k[f]$-module, with quotient field K as $K/k(f)$ is algebraic.

Let P be a place with $f \in \mathcal{O}_P$. As $\mathcal{O}_P$ is a valuation ring, it is integrally closed (3.24.2) and as it contains $k[f]$, also contains B. By (3.31.4), $\mathfrak{B} := \mathfrak{m}_P \cap B$ is then a non-zero prime of B with $\mathcal{O}_P = B_{\mathfrak{B}}$ and $\mathfrak{m}_P = \mathfrak{B}B_{\mathfrak{B}}$; conversely each $B_{\mathfrak{B}}$ is a place, being a discrete valuation ring (hence a valuation ring) with $k \subsetneq B_{\mathfrak{B}} \subsetneq K$ as $f \in B_{\mathfrak{B}} \setminus k$. As $\mathfrak{B}$ is maximal, $B/\mathfrak{B}$ is a field, so $k(\mathfrak{m}_P) = B_{\mathfrak{B}}/\mathfrak{B}B_{\mathfrak{B}} \cong \mathrm{Frac}(B/\mathfrak{B}) = k(\mathfrak{B})$ (3.5.43).

This covers every place P with $f \in \mathcal{O}_P$. For a place P with $f \notin \mathcal{O}_P$ instead, $f^{-1} \in \mathfrak{m}_P \subset \mathcal{O}_P$ (3.24.2), and f^{-1} is transcendental too (else $f = (f^{-1})^{-1}$ would be algebraic); applying the above with f^{-1} in place of f shows $\mathcal{O}_P$ is again a discrete valuation ring. So every place P of K/k is a discrete valuation ring, giving the associated valuation v_P. It vanishes on k, as non-zero elements of k are units of $\mathcal{O}_P$, and $v_P(f) < 0 \iff f \notin \mathcal{O}_P \iff f^{-1} \in \mathfrak{m}_P$ by (3.24.2). ∎

Corollary 6.3.8
Let K/k be an algebraic function field. Then the field of constants $\tilde{k}/k$ has finite degree.

Proof. For any place P we have an embedding $\tilde{k}/k \hookrightarrow k(\mathfrak{m}_P)/k$ which is finite (6.3.6).c). ∎

We prove a number of facts in analogy with algebraic curves.

Lemma 6.3.9 (Every function has a zero)
Let K/k be an algebraic function field and $0 \neq f \in K$. If $f \in \tilde{k}$ (i.e. f is algebraic), then $v_P(f) = 0$ for every place P, so f has no zeros or poles; otherwise (f transcendental) f has between 1 and $[K : k(f)]$ zeros (resp. poles).

Proof. If $f \in \tilde{k}$, then $f \neq 0$ gives $f \in \tilde{k} \setminus \{0\}$, so $f \notin \mathfrak{m}_P$ for every place P (6.3.5), that is $v_P(f) = 0$.

Otherwise, clearly $k[f]$ is a polynomial ring with maximal ideal (f). Therefore by (3.24.12) there is a valuation ring $\mathcal{O}_P \supset k[f]$ such that $\mathfrak{m}_P \cap k[f] = (f)$, which means f has at least one zero.

Recall by (6.3.7) that zeros of f correspond to the maximal ideals of $\overline{k[f]}$ containing f. These are precisely the prime divisors of (f) (see (3.31.13)), so the bound follows from (3.31.20).

The statement about poles of f follows immediately because these are precisely zeros of f^{-1} and $k(f) = k(f^{-1})$. ∎

Proposition 6.3.10 (Places of the Rational Function Field)
Let $K = k(x)$ with x transcendental over k. Then the places of K/k, that is the valuation rings $k \subsetneq \mathcal{O} \subsetneq K$, are exactly the following pairwise distinct discrete valuation rings

a) $\mathcal{O}_{p(x)} := k[x]_{(p(x))}$ where $p \in k[X]$ is an irreducible monic polynomial, with uniformiser $p(x)$ and residue field $k[X]/(p)$, and
b) $\mathcal{O}_\infty := \left\{\frac{f(x)}{g(x)} \;\middle|\; g \neq 0, \deg f \leq \deg g\right\}$, with uniformiser x^{-1} and residue field k.

If k is algebraically closed then the irreducible monic polynomials are the $X - \theta$ with $\theta \in k$, so every place is rational, with uniformiser $x - \theta$ or x^{-1}.

Proof. Let $t := x^{-1}$, which is again transcendental over k with $k(t) = K$.

© David Rufino CC BY-NC-ND 4.0

The listed rings are discrete valuation rings. As $k[x] \cong k[X]$ is a PID (3.15.4), $\mathcal{O}_{p(x)}$ is a discrete valuation ring (3.30.4), with maximal ideal $p(x)\mathcal{O}_{p(x)}$ and residue field $k[x]/(p(x))$ (3.5.27).

Next $\mathcal{O}_\infty = k[t]_{(t)}$: if $f, g \in k[X]$ with $g(0) \neq 0$ and $N := \max(\deg f, \deg g)$ then $f(t)/g(t) = F(x)/G(x)$ with $F(x) := x^N f(1/x)$, $G(x) := x^N g(1/x)$, $\deg F \leq N = \deg G$; conversely $F(x)/G(x)$ with $\deg F \leq \deg G =: N$ equals $f(t)/g(t)$ with $f(t) := t^N F(1/t)$ and $g(t) := t^N G(1/t)$, where $g(0)$ is the leading coefficient of G. So $\mathcal{O}_\infty$ is a discrete valuation ring with uniformiser t and residue field $k[t]/(t) = k$, as above.

Every place is listed. Let $\mathcal{O}$ be a place with maximal ideal $\mathfrak{m}$. Suppose firstly that $x \in \mathcal{O}$. Then $k[x] \subset \mathcal{O}$ and the ideal $k[x] \cap \mathfrak{m} =: (p(x))$ is principal. There is an injective k-algebra homomorphism $k[x]/(p(x)) \hookrightarrow k(\mathfrak{m})$ whence $(p(x))$ is prime (3.3.59). As $\mathcal{O} \neq K$ is not a field, $\mathfrak{m} \neq 0$, and $K = k(x)$ is algebraic over $k[x]$, so $(p(x)) \neq 0$ (3.23.17); we take p monic, and therefore p is irreducible (3.16.11). Suppose $g(x) \notin (p(x))$ then $g(x) \notin \mathfrak{m} \implies g(x)^{-1} \in \mathcal{O}$ (3.24.2). Therefore $\mathcal{O}_{p(x)} \subset \mathcal{O}$, and as $\mathcal{O}_{p(x)}$ is a discrete valuation ring and $\mathcal{O} \neq K$ we conclude that they are equal (3.30.5).

Alternatively suppose that $x \notin \mathcal{O}$ then by (3.24.2) $t = x^{-1} \in \mathfrak{m} \subset \mathcal{O}$. Applying the previous part with t in place of x gives $\mathcal{O} = \mathcal{O}_{p(t)}$ where $\mathfrak{m} \cap k[t] = (p(t))$. As $t \in \mathfrak{m}$ we conclude that $p(t) = t$, and $\mathcal{O} = k[t]_{(t)} = \mathcal{O}_\infty$.

Distinctness. $x \in \mathcal{O}_{p(x)}$ but $x \notin \mathcal{O}_\infty$, as $\deg x > \deg 1$. For distinct monic irreducible $p \neq p'$, $p \notin (p')$, so $p(x)$ is a unit of $\mathcal{O}_{p'(x)}$ but not of $\mathcal{O}_{p(x)}$.

Finally if k is algebraically closed the irreducible monic polynomials are the $X - \theta$ (3.18.58), with residue field $k[X]/(X - \theta) = k$. ∎

Lemma 6.3.11 (Restricting a Place is Proper)
Let L/k' be an algebraic function field, P' a place of L/k', $k \subseteq k'$ with k'/k algebraic, and $K \subseteq L$ a subfield with $\operatorname{trdeg}_k(K) \geq 1$ (e.g. K/k itself an algebraic function field). Then $\mathcal{O}_{P'} \cap K$ is a proper valuation ring of K (and therefore a place.)

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'} \cap K$ is a valuation ring of K (3.24.3). Suppose it equals K, so $K \subset \mathcal{O}_{P'}$. Nonzero elements of K are units of $\mathcal{O}_{P'}$, hence not in $\mathfrak{m}_{P'}$, so $K \to \mathcal{O}_{P'}/\mathfrak{m}_{P'} = k'(\mathfrak{m}_{P'})$ is injective, and $\operatorname{trdeg}_k(K) \leq \operatorname{trdeg}_k(k'(\mathfrak{m}_{P'}))$ (3.18.136). But $k'(\mathfrak{m}_{P'})/k'$ is algebraic (6.3.6).c) and k'/k is algebraic, so the right-hand side is 0, contradicting $\operatorname{trdeg}_k(K) \geq 1$. ∎

Lemma 6.3.12 (Places Restrict Correctly)
Let K/k be an algebraic function field, P a place of K/k, and L/K a field extension. Let $k \subseteq k'' \subset L$ with k''/k algebraic and L/k'' itself an algebraic function field, and P' a place of L/k'' with $\mathcal{O}_P \subset \mathcal{O}_{P'}$. Then $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$; in particular $\mathfrak{m}_P \subset \mathfrak{m}_{P'}$, and $\pi_P \notin k''$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'} \cap K$ is a valuation ring of K containing $\mathcal{O}_P$ (3.24.3), and proper (as K/k is an algebraic function field, so $\operatorname{trdeg}_k(K) = 1 \geq 1$) (6.3.11). By (3.30.5) ($\mathcal{O}_P$ being a DVR, (6.3.7)), a valuation ring of K containing the DVR $\mathcal{O}_P$ is either $\mathcal{O}_P$ itself or all of K; as $\mathcal{O}_{P'} \cap K \neq K$, this forces $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$.

For $0 \neq x \in \mathfrak{m}_P$ we have $x^{-1} \in K \setminus \mathcal{O}_P = K \setminus \mathcal{O}_{P'}$, so $x \in \mathfrak{m}_{P'}$ by (3.24.2): thus $\mathfrak{m}_P \subset \mathfrak{m}_{P'}$. In particular $\pi_P \in \mathfrak{m}_{P'}$ is a non-unit of $\mathcal{O}_{P'}$, so $\pi_P \notin k''$ (nonzero elements of the field $k'' \subset \mathcal{O}_{P'}$ are units). ∎

Proof of (6.3.4). Drafted by Claude Sonnet, reviewed by David Rufino.

A complete regular curve X as above has $k(X)/k$ finitely generated of transcendence degree $\dim X = 1$ (6.1.52), so $k(X)$ is an algebraic function field (6.3.1), in which k is relatively separably closed as X is geometrically integral (6.2.9). The bijection between morphisms and embeddings, together with the degree, is (6.2.120). The map $x_0 \mapsto \mathcal{O}_{X,x}$ is a bijection from X_0 to the valuation rings $k \subsetneq R \subsetneq k(X)$ (6.2.117), which are exactly the places of $k(X)/k$; it is well-defined on X_0 as points of X with the same local ring are identified there (6.2.23).

For the last statement in b), the morphism induces a local homomorphism $\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$, so $\phi_\star\big(\mathcal{O}_{Y,\phi(x)}\big) \subset \mathcal{O}_{X,x}$; if $[y] = [\phi(x)]$ then $\mathcal{O}_{Y,y} = \mathcal{O}_{Y,\phi(x)}$ (6.2.23) and the inclusion follows. Conversely, if $\phi_\star\big(\mathcal{O}_{Y,y}\big) \subset \mathcal{O}_{X,x}$ then $R := \mathcal{O}_{X,x} \cap k(Y)$ is a valuation ring of $k(Y)$ containing the discrete valuation rings $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{Y,\phi(x)}$. It is proper, as ϕ is non-constant: a uniformiser of $\mathcal{O}_{Y,\phi(x)}$ lies in $\mathfrak{m}_{X,x}$ and so is not a unit of R. By (3.30.5) $R = \mathcal{O}_{Y,y} = \mathcal{O}_{Y,\phi(x)}$, so $[y] = [\phi(x)]$ (6.2.23).

It remains to show that every algebraic function field K/k in which k is relatively separably closed is $k(X)$ for such a curve $X =: X_K$.

The affine curves U and U'. Choose $f \in K \setminus \tilde{k}$, which is non-empty as K has transcendence degree 1, so f is transcendental over k, and put $B := \overline{k[f]}^K$ and $B' := \overline{k[f^{-1}]}^K$, subrings of K. These are Dedekind domains with

© David Rufino CC BY-NC-ND 4.0

quotient field K, finite over $k[f]$ and $k[f^{-1}]$ respectively (6.3.7). They are reduced finitely generated k-algebras, so there are affine varieties U and U' with isomorphisms $\iota : \mathcal{O}_U(U) \xrightarrow{\sim} B$ and $\iota' : \mathcal{O}_{U'}(U') \xrightarrow{\sim} B'$ (6.1.36). We write $D(f) \subset U$ for the principal open subset defined by $\iota^{-1}(f)$, and $D(f^{-1}) \subset U'$ for that defined by $\iota'^{-1}(f^{-1})$. These varieties are integral, as $\mathcal{O}_U(U) \cong B$ and $\mathcal{O}_{U'}(U') \cong B'$ are domains (6.2.7), of dimension $\operatorname{trdeg}(\operatorname{Frac}(B)/k) = \operatorname{trdeg}(K/k) = 1$ (6.1.52), and regular: the local ring at a point is a localisation of $\mathcal{O}_U(U)$ (resp. $\mathcal{O}_{U'}(U')$) at a maximal ideal (6.2.24).a), hence a discrete valuation ring, so we may apply (6.2.114).

Since $k(U)$ is a field of fractions of $\mathcal{O}_U(U)$ and of each $\mathcal{O}_U(D(g))$, $g \in \mathcal{O}_U(U)$, via the canonical injections (6.2.18), and $\mathcal{O}_U(D(g)) \cong \mathcal{O}_U(U)_g$ compatibly with restriction (6.2.24).c), the isomorphism ι induces isomorphisms $\iota_g : \mathcal{O}_U(D(g)) \xrightarrow{\sim} B_{\iota(g)}$ and $\iota : k(U) \xrightarrow{\sim} \operatorname{Frac}(B) = K$, all compatible with the inclusions; likewise for ι'.

Gluing. The principal open subsets $D(f)$ and $D(f^{-1})$ are affine, with $\mathcal{O}_U(D(f)) \cong \mathcal{O}_U(U)_f$ and $\mathcal{O}_{U'}(D(f^{-1})) \cong \mathcal{O}_{U'}(U')_{f^{-1}}$ (6.2.24).c), under which the restriction maps are the localisation maps. Taking $g = \iota^{-1}(f)$ and $g' = \iota'^{-1}(f^{-1})$ in the isomorphisms ι_g above we obtain isomorphisms

$$\iota_f : \mathcal{O}_U(D(f)) \xrightarrow{\sim} B[f^{-1}] \quad \text{and} \quad \iota'_{f^{-1}} : \mathcal{O}_{U'}(D(f^{-1})) \xrightarrow{\sim} B'[f]$$

extending ι and ι'. As normalisation commutes with localisation (3.23.9), both $B[f^{-1}]$ and $B'[f]$ are the integral closure of $k[f, f^{-1}]$ in K, so they are equal as subrings of K. Hence $\rho := \iota_f^{-1} \circ \iota'_{f^{-1}} : \mathcal{O}_{U'}(D(f^{-1})) \to \mathcal{O}_U(D(f))$ is an isomorphism of k-algebras. As the functor to reduced finitely generated algebras is full and faithful and reflects isomorphisms (6.2.26), there is a unique isomorphism $\theta : D(f) \xrightarrow{\sim} D(f^{-1})$ with pullback of functions $\theta^\sharp = \rho$, where $\theta^\sharp : \mathcal{O}_{U'}(D(f^{-1})) \xrightarrow{\sim} \mathcal{O}_U(D(f))$, $g \mapsto g \circ \theta$; that is, the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_{U'}(D(f^{-1})) & \xrightarrow[\sim]{\theta^\sharp} & \mathcal{O}_U(D(f)) \\ {\scriptstyle \iota'_{f^{-1}}}\downarrow{\scriptstyle \sim} & & {\scriptstyle \sim}\downarrow{\scriptstyle \iota_f} \\ B'[f] & = & B[f^{-1}] \end{array}$$

As U and U' are not subsets of a common set, we let X_K be the set obtained from $U \sqcup U'$ by identifying each $u \in D(f)$ with $\theta(u)$. Then U and U' inject into X_K with union X_K and intersection $D(f)$, and the transition maps θ and θ^{-1} are regular, so X_K has a unique structure of a variety in which U and U' are open subvarieties (6.2.41).

***X_K* is a regular, geometrically integral curve with function field *K*.** It is irreducible, as U and U' are and they meet in the non-empty set $D(f)$, hence integral, of dimension 1, and regular as it has the regular open cover U, U' (6.2.84). Further $k(X_K) \cong k(U)$ as U is a non-empty open subset of X_K (4.3.16), and $\iota : k(U) \xrightarrow{\sim} K$ as above. As k is relatively separably closed in K, X_K is geometrically integral (6.2.9).

Separatedness. We use (6.2.72): $U \cap U' = D(f)$ is affine, and it remains to show that the map $m : g \otimes g' \mapsto g|_{U\cap U'} \cdot g'|_{U\cap U'}$ on the top row of the following diagram is surjective, where $\mathcal{O}_{X_K}(U) = \mathcal{O}_U(U)$, $\mathcal{O}_{X_K}(U') = \mathcal{O}_{U'}(U')$ and $\mathcal{O}_{X_K}(U \cap U') = \mathcal{O}_U(D(f))$, and where $m' : b \otimes b' \mapsto bb'$ is the k-algebra map induced by the inclusions $B \subset B[f^{-1}]$ and $B' \subset B'[f] = B[f^{-1}]$ of subrings of K.

$$\begin{array}{ccc} \mathcal{O}_{X_K}(U) \otimes_k \mathcal{O}_{X_K}(U') & \xrightarrow{m} & \mathcal{O}_{X_K}(U \cap U') \\ {\scriptstyle \iota\otimes\iota'}\downarrow{\scriptstyle \sim} & & {\scriptstyle \sim}\downarrow{\scriptstyle \iota_f} \\ B \otimes_k B' & \xrightarrow[m']{} & B[f^{-1}] \end{array}$$

It commutes: for $g \in \mathcal{O}_U(U)$ the restriction to $D(f)$ corresponds under ι_f to $\iota(g)$, as ι_f extends ι; and for $g' \in \mathcal{O}_{U'}(U')$ the restriction to $U \cap U'$ is $\theta^\sharp\big(g'|_{D(f^{-1})}\big)$, which ι_f sends to $\iota'_{f^{-1}}\big(g'|_{D(f^{-1})}\big) = \iota'(g')$, as $\iota'_{f^{-1}} = \iota_f \circ \theta^\sharp$. The bottom map m' is surjective, as its image is a subring containing B and $f^{-1} \in B'$, hence $B[f^{-1}]$; so m is surjective.

Points and completeness. Let $\mathbb{P}_K$ be the set of places of K/k. The maximal ideals of B are its non-zero primes, as B is a Dedekind domain, and $\mathfrak{N} \mapsto B_{\mathfrak{N}}$ is a bijection from these to the places containing f (6.3.7); likewise for B' and f^{-1}. Define $\Phi : U \sqcup U' \to \mathbb{P}_K$ by sending a point to its local ring, transported to K: for $u \in U$ this is $\Phi(u) := \iota\big(\mathcal{O}_{U,u}\big)$, where $\mathcal{O}_{U,u}$ is regarded as a subring of $k(U)$ (6.2.18) and $\iota : k(U) \xrightarrow{\sim} K$ is the isomorphism of the set-up, which extends ι_f; for $u' \in U'$ it is $\iota'\big(\mathcal{O}_{U',u'}\big)$ in the same way.

- **Φ is compatible with the identification.** We must show that $\Phi(u) = \Phi(\theta(u))$ for $u \in D(f)$. Write $A := \mathcal{O}_U(D(f))$, $A' := \mathcal{O}_{U'}(D(f^{-1}))$ and $C := B[f^{-1}] = B'[f]$, and let $\mathfrak{M} := \mathfrak{M}_{D(f),u} \subset A$ and $\mathfrak{M}' := \mathfrak{M}_{D(f^{-1}),\theta(u)} \subset$

© David Rufino CC BY-NC-ND 4.0

A' be the maximal ideals of the two points. Then, as subrings of $k(U)$ and $k(U')$, $\mathcal{O}_{U,u} = A_{\mathfrak{M}}$ and $\mathcal{O}_{U',\theta(u)} = A'_{\mathfrak{M}'}$ (6.2.24).a), so

$$\Phi(u) = \iota_f(A_{\mathfrak{M}}) = C_{\iota_f(\mathfrak{M})} \quad \text{and} \quad \Phi(\theta(u)) = \iota'_{f^{-1}}(A'_{\mathfrak{M}'}) = C_{\iota'_{f^{-1}}(\mathfrak{M}')}$$

and it suffices to show $\iota_f(\mathfrak{M}) = \iota'_{f^{-1}}(\mathfrak{M}')$. Now $\left(\theta^\sharp\right)^{-1}(\mathfrak{M}) \subseteq \mathfrak{M}'$ (6.2.25), and both are maximal, so $\mathfrak{M}' = \left(\theta^\sharp\right)^{-1}(\mathfrak{M})$. As $\theta^\sharp = \iota_f^{-1} \circ \iota'_{f^{-1}}$ we have $\iota'_{f^{-1}} = \iota_f \circ \theta^\sharp$, and therefore $\iota'_{f^{-1}}(\mathfrak{M}') = \iota_f\left(\theta^\sharp\left(\left(\theta^\sharp\right)^{-1}(\mathfrak{M})\right)\right) = \iota_f(\mathfrak{M})$. So Φ descends to X_K.

- **Φ is surjective**, as every place contains f or f^{-1} (3.24.2).
- **$\Phi(x) = \Phi(y)$ implies x and y have the same image in $(X_K)_0$.** If $x, y \in U$ the maximal ideals $\mathfrak{M}_{U,x} = \mathfrak{M}_{U,y}$ are recovered from the common local ring, so x and y agree in U_0 (6.2.24).b); likewise in U'. If $x \in U$ and $y \in U'$ have the same place P, then f and f^{-1} are units of $\mathcal{O}_P$, so $x \in D(f)$ and $y \in D(f^{-1})$. Then $\Phi(\theta(x)) = \Phi(x) = \Phi(y)$ with $\theta(x), y \in U'$, so $\theta(x)$ and y agree in U'_0, and $\theta(x) = x$ in X_K.

Therefore $x_0 \mapsto \Phi(x)$ is a bijection from $(X_K)_0$ to $\mathbb{P}_K$, and $\Phi(x) = \iota\left(\mathcal{O}_{X_K,x}\right)$, where $\mathcal{O}_{X_K,x}$ is regarded as a subring of $k(X_K) \cong k(U)$ (4.3.16). For completeness, a proper closed irreducible subset of X_K has dimension 0, so it is the closure of a point (6.2.15), and its function field is the residue field of a point, which is finite over k (6.1.5), so it has no valuation ring containing k other than itself. It remains to take $Z = X_K$ and a valuation ring $k \subset R \subset K$: either $R = K$, which contains every local ring, or R is a place, hence equal to $\iota\left(\mathcal{O}_{X_K,x}\right)$ for the point x corresponding to it. So X_K is complete, and since $k(X_K) \cong K$ and $(X_K)_0$ is in bijection with the set of places, $K \mapsto X_K$ is a quasi-inverse on objects, which completes the proof. ∎

6.3.3 Riemann-Roch Theorem

In what follows we generally assume that $\tilde{k} = k$ (i.e. k is the full constant field of K).

Definition 6.3.13 (Divisors)
Let K/k be an algebraic function field. Denote by $\mathrm{Div}(K)$ *the free abelian group generated by the places of K/k (**divisor group**). The elements are written as a formal sum*

$$D = \sum_P n_P P$$

*with all but finitely many coefficients $n_P = 0$. Define the **degree** homomorphism*

$$\begin{aligned} \mathrm{Div}(K) &\longrightarrow \mathbb{Z} \\ \sum_P n_P P &\longrightarrow \sum_P n_P \cdot [k(\mathfrak{m}_P) : k] \end{aligned}$$

*We may also define $v_P(D) \coloneqq n_P$, and the **support** of D by* $\mathrm{Supp}(D) \coloneqq \{P \mid v_P(D) \neq 0\}$, *a finite set. Finally we may define a partial order*

$$D \leq D' \iff v_P(D) \leq v_P(D') \quad \forall P$$

Definition 6.3.14 (Riemann-Roch Space)
*Let K/k be an algebraic function field and D a divisor. Define the **Riemann-Roch Space** associated to D by*

$$L(D) \coloneqq \{x \in K^\star \mid (x) \geq -D\} \cup \{0\}$$

and define

$$\ell(D) \coloneqq \dim_k L(D)$$

(we shortly show it is a k-vector subspace of K)

Definition 6.3.15 (Principal Divisors)
*Let K/k be an algebraic function field and $x \in K^\star$. Then define the **zero divisor** (resp. **pole divisor**, **principal divisor**) of x as follows*

© David Rufino CC BY-NC-ND 4.0

$$(x)_0 := \sum_{P:v_P(x)>0} v_P(x)P,$$
$$(x)_\infty := \sum_{P:v_P(x)<0} (-v_P(x))P,$$
$$(x) := (x)_0 - (x)_\infty = \sum_P v_P(x)P.$$

which is well-defined by (6.3.9).

Observe similarly $x \in k^\star \iff (x) = 0$. *Define the subgroup of principal divisors by*

$$\mathrm{Princ}(K) := \{(x) \mid x \in K\}$$

and the quotient group $\mathrm{Cl}(K) := \mathrm{Div}(K)/\mathrm{Princ}(K)$ *is called the* ***divisor class group****. We say two divisors* D *and* D' *are* ***linearly equivalent*** *if there exists* $x \in K$ *such that* $D - D' = (x)$, *equivalently if they become equal in the divisor class group. This is written* $D \sim D'$.

We presently prove that these concepts are well-defined, namely every function $x \in K^\star$ has only finitely many zeros. More precisely we can show

Proposition 6.3.16 (Principal Divisors have degree zero)
Let K/k *be an algebraic function field and* $x \in K \setminus \tilde{k}$ *transcendental. Then*

$$\deg((x)_0) = \deg((x)_\infty) = [K : k(x)]$$
$$\deg((x)) = 0$$

In particular there is a well-defined homomorphism

$$\deg : \mathrm{Cl}(K) \longrightarrow \mathbb{Z}$$

Proof. Recall (6.3.7) $k[x]$ is a Dedekind domain, $\overline{k[x]}$ is a finite $k[x]$-module and every place $\mathcal{O}_P$ containing x corresponds to prime ideal $\mathfrak{B}$ of $\overline{k[x]}$. By (3.31.19) then $v_P(x)$ is equal to the ramification index $e_{\mathfrak{B}/(x)}$, and the inertial degree is $f_{\mathfrak{B}/(x)} = [k(\mathfrak{m}_P) : k]$ where (x) is a prime ideal of $k[x]$. From (3.34.61) we have the ramification formula

$$[K : k(x)] = \sum_{\mathfrak{B}|(x)} e_{\mathfrak{B}/(x)} f_{\mathfrak{B}/(x)}$$

Therefore $\deg((x)_0) = [K : k(x)]$ and $\deg((x)_\infty) = \deg\left((x^{-1})_0\right) = [K : k(x^{-1})] = [K : k(x)]$. ∎

Proposition 6.3.17
Let K/k *be an algebraic function field and* D *a divisor. Then*

a) $L(D)$ *is a* k*-vector subspace of* K
b) If $D \sim D'$ *then* $L(D) = L(D')$
c) $L(0) = \tilde{k}$ *so under our standing assumption* $\ell(0) = 1$.
d) If $D < 0$ *then* $L(D) = \{0\}$

Proof.

a) Given $x, y \in L(D)$ then by the strict triangle inequality for valuations $v_P(x+y) \geq \min(v_P(x), v_P(y)) \geq v_P(D)$. Similarly for $\lambda \in k$ we have $v_P(\lambda x) = v_P(\lambda) + v_P(x) = v_P(x)$ (6.3.7).

b) By definition $D' = D + (x)$ we may define a mapping

$$L(D) \longrightarrow L(D')$$
$$y \longrightarrow yx$$

Then

$$v_P(yx) = v_P(x) + v_P(y) \geq v_P(x) + v_P(D) = v_P(D')$$

so this is well-defined. As x^{-1} defines the inverse map it is a bijection.

c) Every non-zero $x \in \tilde{k}$ is a unit of $\mathcal{O}_P$ for each place P (6.3.5), so $v_P(x) = 0$ and $\tilde{k} \subset L(0)$. Suppose $x \in K \setminus \tilde{k}$ then by definition x is transcendental and so is x^{-1}. By (6.3.9) x^{-1} has a zero P, i.e. $v_P(x^{-1}) > 0 \Longrightarrow v_P(x) < 0$. Therefore by definition $x \notin L(0)$ as required.

d) If $x \in \tilde{k}^\star$ then $(x) = 0$ whence $x \notin L(D)$. If $x \in K \setminus \tilde{k}$ then as before x has a pole therefore $x \notin L(D)$ as required.

∎

© David Rufino CC BY-NC-ND 4.0

Lemma 6.3.18
Let K/k be an algebraic function field and $D \leq D'$ divisors. Then $L(D) \subset L(D')$ and

$$\dim_k L(D')/L(D) \leq \deg(D') - \deg(D)$$

In particular $\ell(D)$ is finite and we have

$$\deg(D) - \ell(D) \leq \deg(D') - \ell(D')$$

Proof. Consider a divisor D and suppose $D' \coloneqq D + P$ for some P possibly in the support of D. Let π be a uniformiser for $\mathcal{O}_P$ and define $t = \pi^{v_P(D')}$. We then have a well-defined map

$$\begin{aligned} \psi : L(D') &\longrightarrow k(\mathfrak{m}_P) \\ x &\longrightarrow xt + \mathfrak{m}_P \end{aligned}$$

Observe that $\psi(x) = 0 \iff xt \in \mathfrak{m}_P \iff v_P(xt) > 0 \iff v_P(x) + v_P(D') > 0 \iff v_P(x) \geq -v_P(D) \iff x \in L(D)$. Therefore $\ker(\psi) = L(D)$ and so

$$\dim_k(L(D')/L(D)) \leq [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

Now suppose $D'' = D' + Q$, then by the isomorphism theorem for vector spaces (...)

$$\begin{aligned} \dim_k(L(D'')/L(D)) &= \dim_k(L(D'')/L(D')) + \dim_k(L(D')/L(D)) \\ &\leq \deg(D'') - \deg(D') + \deg(D') - \deg(D) \\ &= \deg(D'') - \deg(D) \end{aligned}$$

from which it follows the general case. ∎

Proposition 6.3.19 (Dimension Upper Bound)
Let K/k be an algebraic function field and D a divisor. Then

$$\ell(D) \leq \deg(D_+) + \dim_k \tilde{k}$$

where D_+ is the positive part of D.

Proof. $L(D) \subset L(D_+)$ so it is enough to show the result for positive divisors. As $L(0) = \tilde{k}$ (...) the result follows from (6.3.18). ∎

We say that the function $\deg(D) - \ell(D)$ is increasing, we also show it is bounded above.

Lemma 6.3.20 (Dimension Lower Bound I)
Let K/k be an algebraic function field. There exists a positive divisor B and integer $\gamma \in \mathbb{Z}$ such that

$$\deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m.

Proof. Let $x \in K$ be transcendental and define $B \coloneqq (x)_\infty$ with $\deg(B) = n$. It's positive by (...). Let $u_1, ..., u_n$ be a $k(x)$-basis for K and a divisor $C \geq 0$ such that $(u_i) \geq -C$. Then we claim that for all integers m the elements $\left\{x^i u_j \;\middle|\; 0 \leq i \leq m, 1 \leq j \leq n\right\}$ are linearly independent over k. But this follows immediately because suppose

$$\sum_{i,j} \lambda_{ij} x^i u_j = 0$$

then

$$\sum_{j=1}^{n} \left(\sum_{i=0}^{m} \lambda_{ij} x^i \right) u_j = 0$$

and by linear independence we have $\lambda_{ij} = 0$.

Further $x^i \in L(iB) \implies x^i \in L(mB)$ and $u_j \in L(C) \implies x^i u_j \in L(mB + C)$. Therefore $\ell(mB + C) \geq n(m+1) = \deg(mB) + n$ for all positive integers m.

On the other hand by (6.3.18) we have $\ell(mB + C) \leq \ell(mB) + \deg(C)$. Therefore

$$\deg(mB) - \ell(mB) \leq \deg(C) - n$$

∎

© David Rufino CC BY-NC-ND 4.0

We may extend this result to all divisors by way of the following lemma.

Lemma 6.3.21
Let K/k be an algebraic function field, B a divisor satisfying the conclusion of (6.3.20) and D an arbitrary divisor. Then there exists divisors D', D'' such that $D \leq D'$, $D' \sim D''$ and $D'' \leq mB$ for some $m \geq 0$.

Proof. Choose a positive divisor $D' \geq D$, then

$$\deg(mB - D') - \ell(mB - D') \leq \deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m. This means

$$\ell(mB - D') \geq m\deg(B) - \deg(D') - \gamma$$

which is positive for sufficiently large m. Therefore we may choose $0 \neq z \in L(mB - D')$. Let $D'' = D' - (z)$. By definition

$$(z) \geq D' - mB \implies D'' \leq mB$$

as required. ∎

Proposition 6.3.22 (Genus is well-defined)
Let K/k be an algebraic function field. Then there exists a positive integer γ such that

$$\deg(D) - \ell(D) \leq \gamma$$

for all divisors D.

Proof. Then using the notation of (6.3.21) we have

$$\begin{aligned}
\deg(D) - \ell(D) &\overset{(6.3.18)}{\leq} && \deg(D') - \ell(D') \\
&\overset{(6.3.16)\ (6.3.17)}{=} && \deg(D'') - \ell(D'') \\
&\overset{(6.3.18)}{\leq} && \deg(mB) - \ell(mB) \\
&\overset{(6.3.20)}{\leq} && \gamma
\end{aligned}$$

∎

Definition 6.3.23 (Genus of a Function Field)
*Let K/k be an algebraic function field with $\tilde{k} = k$. Define the **genus** by*

$$g := \max\{\deg(D) - \ell(D) + 1 \mid D \in \mathrm{Div}(K)\}$$

*Note as $\ell(0) = 1$ the genus g is non-negative. For any divisor $D \in \mathrm{Div}(K)$ define the **index of specialty** as follows*

$$i(D) := \ell(D) - \deg(D) + g - 1$$

which by definition is a non-negative integer. This satisfies a few key properties (6.3.18)

a) $i(0) = g$
b) $i(D) \geq 0$
c) $D \sim D' \implies i(D) = i(D')$
d) $D' \geq D \implies i(D') \leq i(D)$

The Riemann-Roch theorem then takes the following form

$$\ell(D) - i(D) = \deg(D) - g + 1$$

*We say that a divisor D is **non-special** if $i(D) = 0$ (equivalently $\ell(D) = \deg(D) - g + 1$).*

The full Riemann-Roch amounts to an explicit form of $i(D)$. In order to do this we identify it as the dimension of a certain vector space.

Definition 6.3.24 (Repartitions)
*Let K/k be an algebraic function field. Define the K-algebra of **repartitions** of K by*

$$\mathcal{A}_K := \left\{ (\alpha_P)_{P \in \mathbb{P}_K} \;\middle|\; \alpha_P \in K \text{ and } \alpha_P \in \mathcal{O}_P \text{ for all but finitely many places } P \right\}$$

© David Rufino CC BY-NC-ND 4.0

For every divisor $D \in \mathrm{Div}(K)$ define the K-subspace

$$\mathcal{A}_K(D) := \{\alpha \in \mathcal{A}_K \mid v_P(\alpha_P) \geq -v_P(D) \text{ for all } P \in \mathbb{P}_K\}$$

There is a canonical K-algebra homomorphism

$$K \hookrightarrow \mathcal{A}_K$$

which is well-defined by (...). Then clearly we have $K \cap \mathcal{A}_K(D) = L(D)$.

We prove a number of results

Lemma 6.3.25
Let K/k be an algebraic function field. For divisors D, D' such that $D \leq D'$ we have
a) $\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D) = \deg(D') - \deg(D)$
b) $\dim_k((\mathcal{A}_K(D') + K)/(\mathcal{A}_K(D) + K)) = i(D) - i(D')$

Proof. We prove each in turn.

a) By the isomorphism theorem for k-modules it's enough to demonstrate this for the case $D' = D + P$. Let π be a uniformiser for $\mathcal{O}_P$ and let $r := v_P(D) + 1 = v_P(D')$. Then define the k-linear map

$$\begin{aligned} \mathcal{A}_K(D') &\longrightarrow k(\mathfrak{m}_P) \\ \alpha &\longrightarrow (\pi^r \alpha_P)(P) \end{aligned}$$

By definition we have $v_P(\alpha_P) \geq -v_P(D') \implies v_P(\pi^r \alpha_P) \geq 0 \implies \pi^r \alpha_P \in \mathcal{O}_P$ so that this map is well-defined. Further $\pi^r \alpha_P \in \mathfrak{m}_P \iff v_P(\pi^r \alpha_P) > 0 \iff v_P(D) + 1 + v_P(\alpha_P) > 0 \iff v_P(\alpha_P) \geq -v_P(D)$ whence the kernel is precisely $\mathcal{A}_K(D)$. It is clearly surjective because given $f \in \mathcal{O}_P$ we may consider the repartition given by

$$(1, ..., 1, \pi^{-r} f, 1, ...)$$

which clearly maps to f under this map. Therefore by the isomorphism theorem we have

$$\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D) = [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

b) We claim there is an exact sequence of k-modules

$$0 \longrightarrow L(D')/L(D) \overset{\sigma_1}{\to} \mathcal{A}_K(D')/\mathcal{A}_K(D) \overset{\sigma_2}{\to} (\mathcal{A}_K(D') + K)/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

from which the result follows immediately by comparing dimensions. The first map is given by

$$\sigma_1([f]) := (v_P(f))_{P \in \mathbb{P}_K}$$

which is obviously well-defined and injective. The map σ_2 is clearly well-defined and surjective. Further it is evident that $\sigma_2 \circ \sigma_1 = 0$. Therefore it only remains to show that $\ker(\sigma_2) \subset \mathrm{im}(\sigma_1)$. Suppose $\alpha \in \mathcal{A}_K(D')$ then $\sigma_2(\alpha) = 0 \implies \alpha - (f) \in \mathcal{A}_K(D) \implies (f) \in \mathcal{A}_K(D')$ for some $f \in K$. Then by definition this means $f \in L(D')$ and $\sigma_1(f) = \alpha$ as required. ∎

Proposition 6.3.26 (Riemann's Theorem)
*Let K/k be an algebraic function field. Suppose D_0 is **non-special** (there always exists such a divisor) then*
a) $\deg(D) > \ell(D_0) \implies D$ *is non-special*
b) $\mathcal{A}_K = \mathcal{A}_K(D_0) + K$

Proof. We prove each in turn
a) For an arbitrary divisor D we have

$$\begin{aligned} \ell(D - D_0) &= i(D - D_0) + \deg(D - D_0) - g + 1 \\ &\geq 0 + \deg(D) - \ell(D_0) \end{aligned}$$

So when $\deg(D) > \ell(D_0)$ we have $\ell(D - D_0) > 0$. Choose $0 \neq f \in L(D - D_0)$. Then we may define $D' := D + (f) \geq D_0$. Then by (...) $i(D) = i(D') \leq i(D_0) = 0$ whence $i(D) = 0$.

b) Given $\alpha \in \mathcal{A}_K$ choose $D \geq D_0$ such that $\alpha \in \mathcal{A}_K(D)$. Then by (6.3.23) $i(D) \leq i(D_0) = 0$ and therefore $i(D) = 0$. By (6.3.25).b) $\mathcal{A}_K(D) + K = \mathcal{A}_K(D_0) + K \implies \alpha \in \mathcal{A}_K(D_0) + K$ as required.

∎

Proposition 6.3.27 (Explicit form of $i(D)$)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Then

© David Rufino CC BY-NC-ND 4.0

$$i(D) = \dim_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K))$$

Proof. By (6.3.26) there exists some non-special divisor $D' \geq D$ such that $i(D') = 0$ and $\mathcal{A}_K = \mathcal{A}_K(D') + K$. The formula then follows from (6.3.25).b). ∎

Definition 6.3.28 (Weil Divisors)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Define

$$\Omega_K(D) := \{\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k) \mid \mathcal{A}_K(D) + K \subset \ker(\omega)\}$$

As divisors form a directed system we may form the union of ***Weil Divisors***

$$\Omega_K := \bigcup_{D \in \mathrm{Div}(K)} \Omega_K(D) \subset \mathrm{Hom}_k(\mathcal{A}_K, k)$$

Proposition 6.3.29
Let K/k be an algebraic function field. Then there is a canonical isomorphism

$$\mathrm{Hom}_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K), k) \xrightarrow{\sim} \Omega_K(D)$$

In particular $i(D) = \dim_k \Omega_K(D)$. Taking $D = 0$, since $i(0) = g$ (6.3.23) we get

$$\dim_k \Omega_K(0) = g$$

that is, the genus equals the dimension of the space of everywhere-regular Weil differentials.

Proof. This is just the isomorphism theorem for vector spaces. The statement on dimensions follows from (6.3.27) and (3.3.109). ∎

6.3.4 Meromorphic Differentials in the case $k = \mathbb{C}$

In this section we aim to provide a post-hoc motivation for the introduction of Weil Differentials, by showing that they are canonically isomorphic to the classical space of Meromorphic Differentials in the case $k = \mathbb{C}$. This relies on some techniques which are specific to the complex-analytic structure of Riemann Surfaces.

Proposition 6.3.30 (Residue Theorem for $k = \mathbb{C}$)
Let p be a point of a Riemann surface and π a local holomorphic coordinate centered at p. For f meromorphic near p, define

$$\mathrm{Res}_\pi(f) := \mathrm{Res}_p(f\,\mathrm{d}\pi) := \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi$$

for a small loop γ around p – the classical residue of the 1-form $f\,\mathrm{d}\pi$. Writing $g := f \circ \psi$ for the re-expansion of f relative to $\pi' = \psi^{-1}(\pi)$, and $f = \sum_n a_n \pi^n$ for the Laurent series expansion of f at p, then

a) Res_π is k-linear in f;
b) $\mathrm{Res}_\pi(f) = 0$ whenever f is holomorphic at p (i.e. has no pole there);
c) $\mathrm{Res}_\pi(f) = a_{-1}$;
d) $\mathrm{Res}_{\pi'}(g \cdot \psi') = \mathrm{Res}_\pi(f)$;
e) if p ranges over a compact *Riemann surface X and ω is a meromorphic 1-form on X (so has only finitely many poles), then, writing ω_π for the coefficient of ω relative to a local coordinate π at p (so $\omega = \omega_\pi\,\mathrm{d}\pi$ near p, and $\mathrm{Res}_\pi(\omega_\pi) = \mathrm{Res}_p(\omega)$ by definition – independent of the choice of π at each p, by d)),*

$$\sum_p \mathrm{Res}_\pi(\omega_\pi) = 0$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a). Immediate from linearity of the contour integral, $\oint_\gamma (\alpha f + \beta h)\,\mathrm{d}\pi = \alpha \oint_\gamma f\,\mathrm{d}\pi + \beta \oint_\gamma h\,\mathrm{d}\pi$ for $\alpha, \beta \in k$.

b). Immediate from c) below: f holomorphic at p means $a_n = 0$ for every $n < 0$, in particular $a_{-1} = 0$, so $\mathrm{Res}_\pi(f) = a_{-1} = 0$.

c). Term-by-term integration (valid by uniform convergence of the Laurent series on γ) gives $\oint_\gamma f\,\mathrm{d}\pi = \sum_n a_n \oint_\gamma \pi^n\,\mathrm{d}\pi$, and $\oint_\gamma \pi^n\,\mathrm{d}\pi = 2\pi i\delta_{n,-1}$ ($n \neq -1$: exact, $\pi^n\,\mathrm{d}\pi = \mathrm{d}\left(\frac{\pi^{n+1}}{n+1}\right)$, so integrates to 0 around a closed loop; $n = -1$: winding number 1 around $\pi = 0$), so $\mathrm{Res}_\pi(f) = \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi = a_{-1}$.

© David Rufino CC BY-NC-ND 4.0

d). By the chain rule, $f\,\mathrm{d}\pi = f(\psi(\pi'))\psi'(\pi')\,\mathrm{d}\pi' = (g\cdot\psi')\,\mathrm{d}\pi'$: these are the same 1-form ω, written in the two coordinates π and π'. The contour integral $\oint_\gamma \omega$ depends only on ω and γ, not on the coordinate used to write it down – reparametrising γ by π' instead of π changes nothing about which loop is being integrated over – so $\mathrm{Res}_p(\omega)$ computed from either expression gives the same number:

$$\mathrm{Res}_{\pi'}(g\cdot\psi') = \mathrm{Res}_p((g\cdot\psi')\,\mathrm{d}\pi') = \mathrm{Res}_p(\omega) = \mathrm{Res}_p(f\,\mathrm{d}\pi) = \mathrm{Res}_\pi(f)$$

e). This uses two facts special to the compact setting, neither purely local (unlike a)–d) above): ω is **closed** away from its poles ($\omega = f\,\mathrm{d}z$ locally with f holomorphic, so $\mathrm{d}\omega = \frac{\partial f}{\partial \overline{z}}\,\mathrm{d}\overline{z}\wedge\mathrm{d}z = 0$ by the Cauchy–Riemann equations), and X is **compact**. Removing a small disk D_p around each pole leaves $X_0 := X\setminus\bigcup_p D_p$, on which ω is holomorphic, so $\mathrm{d}\omega = 0$ there; Stokes' theorem gives

$$0 = \int_{X_0}\mathrm{d}\omega = \int_{\partial X_0}\omega = -\sum_p \oint_{\partial D_p}\omega = -\sum_p 2\pi i\,\mathrm{Res}_\pi(\omega_\pi)$$

(the sign, as ∂X_0 traverses each ∂D_p oppositely to the usual counterclockwise orientation used to define Res_π, and $\oint_{\partial D_p}\omega = 2\pi i\,\mathrm{Res}_\pi(\omega_\pi)$ by definition), giving $\sum_p \mathrm{Res}_\pi(\omega_\pi) = 0$. ∎

Proposition 6.3.31 (Classical Differentials are Weil Differentials)
Let X be a compact Riemann surface with function field K (so $k=\mathbb{C}$), and let $\mathcal{M}^1(X)$ denote the $\mathbb{C}$-vector space of meromorphic 1-forms on X. For $\omega\in\mathcal{M}^1(X)$, define $\varphi_\omega\in\mathrm{Hom}_k(\mathcal{A}_K,k)$ by

$$\varphi_\omega(\alpha) := \sum_P \mathrm{Res}_\pi(\alpha_P\cdot\omega_\pi),\quad \alpha=(\alpha_P)_P\in\mathcal{A}_K$$

(a finite sum, by b): $\alpha_P\omega_\pi$ is holomorphic, hence contributes 0, at all but finitely many P). Then $\varphi_\omega\in\Omega_K$, and

$$\mathcal{M}^1(X)\longrightarrow\Omega_K,\quad \omega\mapsto\varphi_\omega$$

is a canonical $\mathbb{C}$-linear isomorphism.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\varphi_\omega\in\Omega_K$. Let $D := (\omega)$, the divisor of ω (well-defined: $v_P(\omega_\pi)$ is independent of the choice of π at P, as $\omega_{\pi'} = (\omega_\pi\circ\psi)\cdot\psi'$ by the chain rule – the same identity underlying d) – and ψ' is a unit, so composition with ψ and multiplication by ψ' both preserve order of vanishing). We show $\mathcal{A}_K(D)+K\subset\ker(\varphi_\omega)$. For $\alpha\in\mathcal{A}_K(D)$, $v_P(\alpha_P)\geq -v_P(D) = -v_P(\omega_\pi)$ for every P, so $v_P(\alpha_P\omega_\pi)\geq 0$: each $\alpha_P\omega_\pi$ is holomorphic at P, so $\mathrm{Res}_\pi(\alpha_P\omega_\pi)=0$ by b), giving $\varphi_\omega(\alpha)=0$. For $f\in K$ (embedded diagonally, $\alpha_P=f$ for every P), $\varphi_\omega(f,f,\ldots)=\sum_P\mathrm{Res}_\pi(f\cdot\omega_\pi)=\sum_P\mathrm{Res}_\pi((f\omega)_\pi)=0$ by e) (the Residue Theorem), applied to the meromorphic 1-form $f\omega\in\mathcal{M}^1(X)$. So φ_ω vanishes on $\mathcal{A}_K(D)+K$, that is $\varphi_\omega\in\Omega_K(D)\subset\Omega_K$.

$\mathbb{C}$-linearity. Immediate from a): $\varphi_{\lambda\omega_1+\mu\omega_2}(\alpha)=\sum_P\mathrm{Res}_\pi\big(\alpha_P(\lambda\omega_1+\mu\omega_2)_\pi\big)=\lambda\varphi_{\omega_1}(\alpha)+\mu\varphi_{\omega_2}(\alpha)$ for $\lambda,\mu\in\mathbb{C}$, using that $(\lambda\omega_1+\mu\omega_2)_\pi=\lambda(\omega_1)_\pi+\mu(\omega_2)_\pi$ (coefficients relative to a fixed π add linearly) and linearity of Res_π in its argument.

Injectivity. Suppose $\varphi_\omega=0$; we show $\omega=0$. Fix P and, for $n\in\mathbb{Z}$, let $\alpha^{(n)}\in\mathcal{A}_K$ be the repartition with $\alpha_P^{(n)} := \pi^{-n}$ and $\alpha_Q^{(n)} := 0$ for $Q\neq P$ (a valid repartition: it is holomorphic away from the single place P). Writing $\omega_\pi = \sum_m a_m\pi^m$, $0=\varphi_\omega(\alpha^{(n)}) = \mathrm{Res}_\pi(\pi^{-n}\omega_\pi) = a_{n-1}$ by c). As n was arbitrary, every coefficient of ω_π vanishes, so ω is 0 near P; as P was arbitrary, $\omega\equiv 0$ on X.

Surjectivity. *Sketch.* This needs the classical Mittag-Leffler theorem for meromorphic differentials: given principal parts (polar Laurent tails) prescribed at finitely many points of X, satisfying the residue-sum-zero constraint of e), there exists a global meromorphic 1-form on X realising them – solved via the $\overline{\partial}$-equation (Hodge theory / finite-dimensionality of $H^1(X,\mathcal{O})$), genuinely global analytic content with no counterpart among a)–e) above. Given $\psi\in\Omega_K(D)$, read off, at each pole P of D, the local principal part ψ demands by testing against the single-place repartitions $\alpha_P^{(n)}=\pi^{-n}$ used in the injectivity argument above; since ψ vanishes on K by construction, these prescribed principal parts automatically satisfy the residue-sum-zero compatibility condition Mittag-Leffler requires, and it produces a global ω realising them, with $\varphi_\omega=\psi$. ∎

6.3.5 Derivations on the Function Field

Proposition 6.3.32 (A Distinguished Basis for $\mathrm{Der}_k(K,K)$)
Let $x\in K$ be transcendental over k and assume k is perfect. Then there is a unique $\partial_x\in\mathrm{Der}_k(K,K)$ with $\partial_x(x)=1$, and $D=D(x)\cdot\partial_x$ for every $D\in\mathrm{Der}_k(K,K)$.

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.34.44), $\dim_K \mathrm{Der}_k(K) = 1$. So there is at least one non-zero D_0. If $D_0(x) = 0$ then it vanishes on $k(x)$ by (3.25.9), (3.25.10) and therefore on K as $K/k(x)$ is separable (3.18.103) (3.34.42), a contradiction. Therefore $D_0(x) \neq 0$ for any non-zero derivation D_0. Therefore define $\partial_x := D_0(x)^{-1}D$ and by monogenicity we are done. ∎

Proposition 6.3.33 (Uniformisers are Transcendental)
Let P be a place of K/k, π a uniformiser. Then π is transcendental over k.

Proof. By (...) $k \cap \mathfrak{m}_P = (0)$ whence $\pi \notin k$. By assumption $\tilde{k} = k$ whence π is transcendental. ∎

Proposition 6.3.34 (Chain Rule for Uniformiser Derivations)
Let $x, x' \in K$ be transcendental. Then

$$\partial_{x'}(t) = \partial_{x'}(x) \cdot \partial_x(t) \quad \text{for every } t \in K.$$

Proof. This is immediate from (6.3.32). ∎

6.3.6 Residue Symbol at a Place

The residue symbol $\mathrm{Res}_P(z\,\mathrm{d}t)$ built in this subsection is defined via the Laurent coefficients of $z\partial_\pi(t)$ relative to a choice of uniformiser π, and shown independent of that choice (6.3.43) – exactly the invariance a genuine differential-form residue ought to have. This invariance is not straightforward in characteristic p: it reduces (6.3.43) to computing $(\psi^n\psi')_{-1}$ for the change-of-uniformiser series ψ, an identity which the naive derivative-of-a-power argument only settles when $n+1$ is invertible – automatic in characteristic 0, but not when $p \mid (n+1)$. We settle it uniformly in every characteristic by proving the identity once, universally, over $\mathbb{Z}[a_1, a_1^{-1}, a_2, a_3, ...]$ (indeterminate coefficients, torsion-free over $\mathbb{Z}$) and specialising (6.3.38) – the same universal-$\mathbb{Z}$-polynomial philosophy used, e.g., in [Sti09] (Chap. 4, Prop. 4.2.9); another route entirely avoiding a characteristic-p detour is via finite-potent operators [Tat68] (§§1–2).

Throughout this subsection k is perfect, so the residue field $k(P)$ of every place P of K/k is a finite separable extension of k (3.18.103) and the coefficient expansion (3.32.17) embeds K in $k(P)((X))$. The residue of $z\,\mathrm{d}t$ at P is first a $k(P)$-valued Laurent coefficient, and the residue symbol $\mathrm{Res}_P(z\,\mathrm{d}t) \in k$ is its image under the trace $\mathrm{Tr}_{k(P)/k}$. At a *rational* place, that is $k(P) = k$, the trace is the identity and this is the residue of the classical case, a k-valued Laurent coefficient.

The map L_π below is the algebraic analogue of the Laurent expansion $f = \sum_{j \geq -N} a_j (z-p)^j$ of a meromorphic function f at a point p of a Riemann surface, in a local coordinate $z - p$ (the case $k = \mathbb{C}$ and P a rational place). Here K plays the role of the meromorphic functions on the curve, $\mathcal{O}_P$ of those holomorphic at P and the uniformiser π of the local coordinate, and the coefficients lie in the residue field $k(P)$, which is $\mathbb{C}$ at every point of a Riemann surface. The residue of a differential $z\,\mathrm{d}t$ at P will be the trace $\mathrm{Tr}_{k(P)/k}$ of the coefficient $a_{-1} \in k(P)$ of the expansion of $z\partial_\pi(t)$, so that it lands in k; at a rational place this is a_{-1} itself, as in the classical case.

Proposition 6.3.35 (Laurent Coefficients at a Place)
Let P be a place of K/k, π a uniformiser of $\mathcal{O}_P$ and $k(P) := \mathcal{O}_P/\mathfrak{m}_P$ (a finite separable extension of k, as k is perfect). For $u \in \mathcal{O}_P$ write $[u]_N$ for its class in $\mathcal{O}_P/\mathfrak{m}_P^{N+1}$, and let $s_N : k(P) \to \mathcal{O}_P/\mathfrak{m}_P^{N+1}$ be the k-algebra sections of (3.25.18), which do not depend on π. There is a unique k-algebra homomorphism

$$L_\pi : K \to k(P)((X))$$

with $L_\pi(\pi) = X$ such that

- *L_π preserves valuation: $v_X(L_\pi(z)) = v_P(z)$ for every $z \in K^\star$ (v_X being the valuation of the DVR $k(P)[[X]]$ (3.32.16));*
- *L_π induces the identity on the residue field: $L_\pi(u) \equiv \overline{u} \pmod{X}$ for $u \in \mathcal{O}_P$.*

It has the following further properties.

a) L_π is injective, and $z \in \mathcal{O}_P \iff L_\pi(z) \in k(P)[[X]]$ for every $z \in K$.

*b) For $z \in K^\star$, $L_\pi(z) = \sum_{j \geq v_P(z)} a_j X^j$ with $a_{v_P(z)} \neq 0$ (the **Laurent coefficients** $a_j \in k(P)$ of z at P relative to π).*

c) If $u \in \mathcal{O}_P$ and $L_\pi(u) = \sum_i c_i X^i$, then for every $N \geq 0$

$$[u]_N = \sum_{i \leq N} s_N(c_i)[\pi^i]_N.$$

d) For $c \in k(P)$ and $c' \in \mathcal{O}_P$ with $[c']_N = s_N(c)$ we have $L_\pi(c') \equiv c \pmod{X^{N+1}}$, for every uniformiser π.

If P is rational then $k(P) = k$, the s_N are induced by $k \subseteq \mathcal{O}_P$ and $L_\pi : K \to k((X))$.

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $k(P)/k$ is finite (6.3.6).c) and k is perfect, $k(P)/k$ is separable algebraic (3.18.103), so (3.32.17) applies to $A := \mathcal{O}_P$, with the sections s_N of (3.25.18). It gives an injective k-algebra homomorphism $L_\pi : K \to k(P)((X))$ with $L_\pi(\pi) = X$ which on $\mathcal{O}_P$ is given by the **digits** of the π-adic expansion, that is the unique coefficients $c_i(u) \in k(P)$ of (3.32.17) with $[u]_N = \sum_{i \le N} s_N(c_i(u))[\pi^i]_N$ for every N, so that $L_\pi(u) = \sum_i c_i(u) X^i$.

Valuation and residue. $v_X(L_\pi(u)) = v_P(u)$ for $0 \neq u \in \mathcal{O}_P$ is the valuation formula of that proposition, and for $z = a/s \in K^\star$ $(a, s \in \mathcal{O}_P)$, $L_\pi(z) = L_\pi(a) L_\pi(s)^{-1}$, so $v_X(L_\pi(z)) = v_P(a) - v_P(s) = v_P(z)$ (multiplicativity of v_X (3.32.15)). In particular $z \in \mathcal{O}_P \iff v_P(z) \ge 0 \iff L_\pi(z) \in k(P)[[X]]$, and $a_j = 0$ for $j < v_P(z)$ with $a_{v_P(z)} \neq 0$. For $u \in \mathcal{O}_P$ the case $N = 0$ of the displayed identity reads $[u]_0 = s_0(c_0(u)) = c_0(u)$ (s_0 is the identity of $k(P)$), that is $c_0(u) = \overline{u}$: L_π induces the identity on residue fields.

Compatibility with the sections. Let $c \in k(P)$ and $c' \in \mathcal{O}_P$ with $[c']_N = s_N(c)$. Then $(c, 0, ..., 0)$ satisfies the defining identity for c' at level N, so by uniqueness of the digits $c_0(c') = c$ and $c_i(c') = 0$ for $1 \le i \le N$, that is $L_\pi(c') \equiv c \pmod{X^{N+1}}$. The s_N are unique, hence independent of π.

Uniqueness. Let $\phi : K \to k(P)((X))$ be a valuation-preserving k-algebra homomorphism with $\phi(\pi) = X$ inducing the identity on the residue field. For $N \ge 0$ it maps $\mathcal{O}_P$ into $k(P)[[X]]$ and $\mathfrak{m}_P^{N+1}$ into $X^{N+1} k(P)[[X]]$, so induces a k-algebra homomorphism $\phi_N : \mathcal{O}_P/\mathfrak{m}_P^{N+1} \to k(P)[X]/(X^{N+1})$. Then $\phi_N \circ s_N : k(P) \to k(P)[X]/(X^{N+1})$ is a k-algebra homomorphism which reduces to the identity of $k(P)$ modulo X (as ϕ induces the identity on residue fields), as does the inclusion of the constants. By the uniqueness of such sections (3.25.18) (applied to the local k-algebra $k(P)[X]/(X^{N+1})$) they agree: $\phi_N(s_N(c)) = c$. So for $u \in \mathcal{O}_P$ with digits c_i,

$$\phi(u) \equiv \sum_{i \le N} \phi_N(s_N(c_i)) X^i = \sum_{i \le N} c_i X^i \equiv L_\pi(u) \pmod{X^{N+1}}$$

for every N, so $\phi = L_\pi$ on $\mathcal{O}_P$ and hence on $K = \mathrm{Frac}(\mathcal{O}_P)$. ∎

Definition 6.3.36 (Residue Symbol of a Laurent Series)
Let A be a commutative ring, π an indeterminate, and $z, t \in A((\pi))$. Write ∂_π for the formal derivative, $\partial_\pi\big(\sum_n a_n \pi^n\big) := \sum_n n a_n \pi^{n-1}$, and define

$$\mathrm{Res}_\pi(z\,\mathrm{d}t) := (z \partial_\pi(t))_{-1} \in A,$$

the coefficient of π^{-1} in $z\partial_\pi(t)$.

Definition 6.3.37 (Residue Symbol at a Place)
Let P be a place of K/k, π a uniformiser, and $t, z \in K$. Let $L_\pi : K \hookrightarrow k(P)((X))$ be the Laurent coefficient embedding (6.3.35). Define the $k(P)$-valued residue

$$\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) := \mathrm{Res}_\pi(L_\pi(z)\,\mathrm{d}L_\pi(t)) \in k(P),$$

*where Res_π is the residue of (6.3.36) with $A := k(P)$, and $\mathrm{d}L_\pi(t) := \partial_\pi(L_\pi(t))\,\mathrm{d}\pi$, and the **residue symbol***

$$\mathrm{Res}_P(z\,\mathrm{d}t) := \mathrm{Tr}_{k(P)/k}\Big(\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big) \in k.$$

Since L_π commutes with the uniformiser derivation, $L_\pi(\partial_\pi(t)) = \partial_\pi(L_\pi(t))$ (6.3.41), one has explicitly

$$\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) = L_\pi(z\partial_\pi(t))_{-1} = (L_\pi(z)\partial_\pi(L_\pi(t)))_{-1}.$$

If P is rational then $k(P) = k$, the trace is the identity and $\mathrm{Res}_P = \mathrm{Res}_P^k$ is the k-valued Laurent coefficient. We show that both are independent of the choice of uniformiser π below (6.3.43).

Lemma 6.3.38 (Residue of $\psi^n\psi'$)
Let F be a field and $\psi \in XF[[X]]$ with $\psi'(0) \neq 0$ (equivalently, ψ has order exactly 1). For every $n \in \mathbb{Z}$,

$$(\psi^n \psi')_{-1} = \delta_{n,-1}$$

Remark 6.3.39
The simplest proof using the identity $(n+1)\psi^n\psi' = (\psi^{n+1})'$ fails when $\mathrm{char}(k) = p \mid n+1$, so some additional work is required.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

© David Rufino CC BY-NC-ND 4.0

$n \geq 0$. $\psi \in Xk[[X]]$ has no negative powers, so neither do ψ^n or ψ'; their product has no X^{-1} term, giving 0, matching $\delta_{n,-1} = 0$.

$n = -1$. Write $\psi(X) = Xv(X)$ with $v \in k[[X]]^\star$ a unit ($v(0) = \psi'(0) \neq 0$). Then $\psi' = v + Xv'$, so

$$\psi^{-1}\psi' = \frac{v + Xv'}{Xv} = \frac{1}{X} + \frac{v'}{v}$$

The second term is a genuine power series (product of v' and the unit v^{-1}, both in $k[[X]]$), so has zero -1-coefficient; the first contributes $\left(X^{-1}\right)_{-1} = 1$. Total: 1, matching $\delta_{-1,-1}$.

$n \leq -2$: a universal identity. Fix $n \leq -2$; we prove the identity over a universal coefficient ring first, then specialise (see the remark above for why). Let $R := \mathbb{Z}[a_1, a_2, a_3, ...][a_1^{-1}]$, the polynomial ring over $\mathbb{Z}$ in countably many indeterminates $a_1, a_2, ...$, localised to invert a_1 – an integral domain of characteristic 0 (a localisation of the domain $\mathbb{Z}[a_1, a_2, ...]$, itself a domain as a polynomial ring over the domain $\mathbb{Z}$, (3.7.3), applied to finitely many variables at a time since any product only ever involves finitely many of them). Let $\Psi := \sum_{i\geq 1} a_i X^i \in XR[[X]]$, so $\Psi = a_1 X \cdot U$ for $U := 1 + \sum_{i\geq 2}(a_i/a_1)X^{i-1} \in 1 + XR[[X]]$; as U has constant coefficient $1 \in R^\star$, (3.32.13) gives $U \in R[[X]]^\star$, and hence $\Psi \in R((X))^\star$ too ($a_1 \in R^\star$, and X a unit of $R((X))$ directly by construction of the formal Laurent series ring, (3.32.1)). So $\Psi^n\Psi' \in R((X))$ is well-defined, with a well-defined coefficient of X^{-1}, which we still write $(\Psi^n\Psi')_{-1} \in R$.

As before (the power rule and vanishing of a derivative's X^{-1}-coefficient being valid over any commutative ring, not just k), $(n+1)\Psi^n\Psi' = \left(\Psi^{n+1}\right)'$ (3.32.20) (Ψ being a unit) has vanishing X^{-1}-coefficient:

$$(n+1)\cdot(\Psi^n\Psi')_{-1} = \left(\left(\Psi^{n+1}\right)'\right)_{-1} = 0 \quad \text{in } R$$

As $n \leq -2$, $n+1$ is a nonzero integer; its image in R is therefore nonzero, R having characteristic 0. So $(n+1)\cdot(\Psi^n\Psi')_{-1} = 0$ forces $(\Psi^n\Psi')_{-1} = 0$ in R, by cancellation in the domain R.

Specialising. Write $\psi = \sum_{i\geq 1} c_i X^i$, so $c_1 = \psi'(0) \neq 0$ by hypothesis. As $c_1 \in k^\star$, the assignment $a_1 \mapsto c_1$, $a_i \mapsto c_i$ ($i \geq 2$) extends, by the universal property of localisation (3.5.5) (applied to $\mathbb{Z}[a_1, a_2, ...] \to k$, $a_1 \mapsto c_1$ a unit), to a ring homomorphism $\phi : R \to k$. Applying ϕ coefficientwise gives a ring homomorphism $\hat{\phi} : R((X)) \to k((X))$ (directly from the convolution formula for products, (3.32.1)) with $\hat{\phi}(\Psi) = \psi$, compatible with the formal derivative ($\hat{\phi}(f)' = \hat{\phi}(f')$, immediate from the coefficientwise formulas $f' = \sum_m mc_m X^{m-1}$ and $\phi(mc_m) = m\phi(c_m)$) and with the coefficient of X^{-1} (literally the same index, carried along by ϕ). So

$$(\psi^n\psi')_{-1} = \hat{\phi}(\Psi^n\Psi')_{-1} = \phi\Big((\Psi^n\Psi')_{-1}\Big) = \phi(0) = 0$$

using $(\Psi^n\Psi')_{-1} = 0$ in R (just shown), matching $\delta_{n,-1} = 0$ for $n \leq -2$. ∎

Lemma 6.3.40 (Laurent Expansions Compose Across Uniformisers)
Let P be a place of K/k, π, π' uniformisers at P, and $\psi := L_{\pi'}(\pi) \in Xk(P)[[X]]$.

Then for every $w \in K$,

$$L_{\pi'}(w) = L_\pi(w) \circ \psi$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We may assume $w \neq 0$. Write $f := L_\pi(w) = \sum_j a_j X^j$ and $m := v_P(w)$, so $a_j = 0$ for $j < m$. As $v(\psi) = 1$ and the order function is multiplicative on the field $k(P)((X))$ (3.32.15), $v(\psi^j) = j$ for every j. So the family $h_i := a_{m+i}\psi^{m+i}$ ($i \geq 0$) is summable (3.32.9), as $v(h_i) \geq m + i$, and by definition of the substitution (3.32.21) $f \circ \psi = \sum_{j\geq m} a_j\psi^j = \sum_{i\geq 0} h_i$. By the Convergence Criterion (3.32.12) it suffices to show that for every $n \geq m$

$$L_{\pi'}(w) - \sum_{j=m}^{n} a_j\psi^j \in X^{n+1}k(P)[[X]], \qquad (\star)$$

since then the partial sums $\sum_{i\leq n-m} h_i$ converge to $L_{\pi'}(w)$, and so $L_{\pi'}(w) = \sum_i h_i = f \circ \psi$.

Proof of ($\star$) for a rational place. Here $k(P) = k \subseteq \mathcal{O}_P$, so $a_j \in k \subseteq \mathcal{O}_P$. Write $w = \pi^m u$ with $u \in \mathcal{O}_P^\star$; the digits of u are $c_i = a_{m+i}$ (6.3.35), so for $n \geq m$

$$w - \sum_{j=m}^{n} a_j\pi^j = \pi^m\left(u - \sum_{i\leq n-m} a_{m+i}\pi^i\right) \in \pi^m\mathfrak{m}_P^{n-m+1} = \mathfrak{m}_P^{n+1}.$$

© David Rufino CC BY-NC-ND 4.0

Apply $L_{\pi'}$: it preserves valuation, so the left side maps into $X^{n+1}k[[X]]$ (6.3.35); it is a k-algebra homomorphism of fields with $L_{\pi'}(\pi)=\psi$, so it sends $\pi^j\mapsto\psi^j$ for every $j\in\mathbb{Z}$ (3.2.6) and fixes each $a_j\in k$. Hence $L_{\pi'}(w)-\sum_{j=m}^{n}a_j\psi^j\in X^{n+1}k[[X]]$, which is $(\star)$.

Proof of $(\star)$ for a general place. The coefficients $a_j\in k(P)$ need not lie in $\mathcal{O}_P$, so we replace them by lifts through the sections s_N of (6.3.35). Fix $n\geq m$ and put $N:=n-m$. Write $w=\pi^m u$ with $u\in\mathcal{O}_P^\star$, with digits $c_i=a_{m+i}$, and choose $\tilde{a}_j\in\mathcal{O}_P$ with $\left[\tilde{a}_j\right]_N=s_N(a_j)$ for $m\leq j\leq n$. Then $[u]_N=\sum_{i\leq N}s_N(c_i)[\pi^i]_N=\left[\sum_{i\leq N}\tilde{a}_{m+i}\pi^i\right]_N$ gives, as before,

$$w-\sum_{j=m}^{n}\tilde{a}_j\pi^j=\pi^m\left(u-\sum_{i\leq N}\tilde{a}_{m+i}\pi^i\right)\in\pi^m\mathfrak{m}_P^{N+1}=\mathfrak{m}_P^{n+1}.$$

Applying $L_{\pi'}$ exactly as in the rational case,

$$L_{\pi'}(w)-\sum_{j=m}^{n}L_{\pi'}(\tilde{a}_j)\psi^j\in X^{n+1}k(P)[[X]].$$

It remains to replace $L_{\pi'}(\tilde{a}_j)$ by a_j. By the compatibility with the sections (6.3.35) (valid for every uniformiser, in particular π') we have $L_{\pi'}(\tilde{a}_j)\equiv a_j\pmod{X^{N+1}}$, and $v(\psi^j)=j\geq m$, so $\left(L_{\pi'}(\tilde{a}_j)-a_j\right)\psi^j\in X^{N+1+j}k(P)[[X]]\subseteq X^{n+1}k(P)[[X]]$. This gives $(\star)$. ∎

Corollary 6.3.41 (Laurent Coefficients of the Uniformiser Derivation)
Let P be a place of K/k, π a uniformiser. For every $w\in K$,

$$L_\pi(\partial_\pi(w))=\frac{\mathrm{d}}{\mathrm{d}X}L_\pi(w)$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $M:=k(P)((X))$, the target of $L_\pi:K\hookrightarrow k(P)((X))$ (6.3.35), viewed as a K-module through L_π, a k-algebra homomorphism (6.3.35). By (3.34.44) (with $n=1$, transcendence basis π (6.3.33)), $\mathrm{Der}_k(K,M)\cong M$ via $D\mapsto D(\pi)$, so a k-derivation $K\to M$ is uniquely determined by its value at π.

Both $D_1:=L_\pi\circ\partial_\pi$ and $D_2:=\frac{\mathrm{d}}{\mathrm{d}X}\circ L_\pi$ are k-derivations $K\to M$: for D_1, $L_\pi(\partial_\pi(xy))=L_\pi(x\partial_\pi(y)+y\partial_\pi(x))=x\cdot D_1(y)+y\cdot D_1(x)$, using additivity and multiplicativity of L_π; for D_2, the same identity holds by the Leibniz rule for formal differentiation together with $L_\pi(xy)=L_\pi(x)L_\pi(y)$; and $D_1(\lambda)=L_\pi(\partial_\pi(\lambda))=0$, $D_2(\lambda)=0$ for $\lambda\in k$, as $L_\pi(\lambda)$ is a constant of $k(P)((X))$. Both send $\pi\mapsto 1$ ($L_\pi(\partial_\pi(\pi))=L_\pi(1)=1$, $\frac{\mathrm{d}}{\mathrm{d}X}(L_\pi(\pi))=\frac{\mathrm{d}}{\mathrm{d}X}(X)=1$), so $D_1=D_2$, that is $L_\pi(\partial_\pi(w))=\frac{\mathrm{d}}{\mathrm{d}X}L_\pi(w)$ for every $w\in K$. ∎

Proposition 6.3.42 (The Uniformiser Derivation is Regular)
Let P be a place of K/k, π a uniformiser. Then $\partial_\pi(\mathcal{O}_P)\subseteq\mathcal{O}_P$.

Proof. This essentially follows from (6.3.35) and the fact that $\frac{\mathrm{d}}{\mathrm{d}X}$ preserves $k(P)[[X]]$. ∎

Corollary 6.3.43 (Well-Definedness of the Residue Symbol at a Place)
Let P be a place of K/k, π,π' uniformisers at P, and $z,t\in K$. Then

$$L_\pi(z\partial_\pi(t))_{-1}=L_{\pi'}(z\partial_{\pi'}(t))_{-1}$$

an equality in $k(P)$; that is, $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)$, and hence $\mathrm{Res}_P(z\,\mathrm{d}t)$, of (6.3.37) does not depend on the choice of uniformiser π used to compute it.

Proof. Write $w:=z\partial_\pi(t)$ and $\psi:=L_{\pi'}(\pi)\in Xk(P)[[X]]$.

$\psi'(0)\neq 0$. Writing $\psi=\sum_j a_jX^j$ ($a_j\in k(P)$), this is the Laurent expansion of $\pi\in K^\star$ relative to π' (6.3.35), so $a_j=0$ for $j<v_P(\pi)$ and $a_{v_P(\pi)}\neq 0$; as $v_P(\pi)=1$ (π is itself a uniformiser), $a_0=0$ and $a_1\neq 0$. The constant term of $\psi'=\sum_j ja_jX^{j-1}$ is its $j=1$ term, $1\cdot a_1=a_1\neq 0$, that is $\psi'(0)\neq 0$.

$L_\pi(w)_{-1}=(L_{\pi'}(w)\cdot\psi')_{-1}$. Write $f:=L_\pi(w)$ and $g:=f\circ\psi$ (well-defined, (3.32.21), as ψ has invertible linear term, just shown). Both sides of $f_{-1}=(g\cdot\psi')_{-1}$ are $k(P)$-linear in f (composition with ψ, multiplication by ψ', and $(-)_{-1}$ are all $k(P)$-linear), so it suffices to check $f=X^n$, $n\in\mathbb{Z}$, that is $(\psi^n\psi')_{-1}=\delta_{n,-1}$ – exactly (6.3.38). By (6.3.40), $g=L_\pi(w)\circ\psi=L_{\pi'}(w)$, giving $L_\pi(w)_{-1}=f_{-1}=(g\cdot\psi')_{-1}=(L_{\pi'}(w)\cdot\psi')_{-1}$.

By (6.3.41) (applied relative to π'), $L_{\pi'}(\partial_{\pi'}(\pi))=\frac{\mathrm{d}}{\mathrm{d}X}L_{\pi'}(\pi)=\psi'$. As $L_{\pi'}$ is a ring homomorphism (6.3.35),

$$L_\pi(z\partial_\pi(t))_{-1}=L_\pi(w)_{-1}=(L_{\pi'}(w)\cdot L_{\pi'}(\partial_{\pi'}(\pi)))_{-1}=(L_{\pi'}(w\partial_{\pi'}(\pi)))_{-1}.$$

© David Rufino CC BY-NC-ND 4.0

By (6.3.34), $\partial_\pi(t)\partial_{\pi'}(\pi) = \partial_{\pi'}(t)$, so $w\partial_{\pi'}(\pi) = z\partial_\pi(t)\partial_{\pi'}(\pi) = z\partial_{\pi'}(t)$, giving $L_\pi(z\partial_\pi(t))_{-1} = L_{\pi'}(z\partial_{\pi'}(t))_{-1}$. ∎

Lemma 6.3.44 (Residues are Invariant under Isomorphism)
Let $\varphi : K \to K'$ be an isomorphism of k-algebras between algebraic function fields K/k, K'/k, P a place of K/k and P' the place of K'/k with $\mathcal{O}_{P'} = \varphi(\mathcal{O}_P)$. Then φ induces an isomorphism $\overline{\varphi} : k(P) \to k(P')$ of k-algebras, and for all $z, t \in K$

$$\operatorname{Res}_{P'}^{k(P')}(\varphi(z)\,\mathrm{d}\varphi(t)) = \overline{\varphi}\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big), \qquad \operatorname{Res}_{P'}(\varphi(z)\,\mathrm{d}\varphi(t)) = \operatorname{Res}_P(z\,\mathrm{d}t).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix a uniformiser π at P. Then $\pi' := \varphi(\pi)$ is a uniformiser at P', and φ preserves valuations. Write $L := L_\pi : K \to k(P)((X))$ and $L' := L_{\pi'} : K' \to k(P')((X))$, and $\overline{\varphi}_*$ for the isomorphism $k(P)((X)) \to k(P')((X))$ applying $\overline{\varphi}$ to the coefficients. We claim that $L' \circ \varphi = \overline{\varphi}_* \circ L$. Indeed $\Psi := (\overline{\varphi}_*)^{-1} \circ L' \circ \varphi : K \to k(P)((X))$ is a k-algebra homomorphism with $\Psi(\pi) = X$ which preserves valuation, as each factor does, and induces the identity on the residue field: for $u \in \mathcal{O}_P$, $L'(\varphi(u)) \equiv \overline{\varphi(u)} = \overline{\varphi}(\overline{u}) \pmod{X}$. So $\Psi = L$ by the uniqueness in (6.3.35), which is the claim.

Next, $\varphi \circ \partial_\pi \circ \varphi^{-1}$ is a k-derivation of K' taking π' to 1, so it equals $\partial_{\pi'}$ by (6.3.32), as π' is transcendental over k (6.3.33). Hence

$$\operatorname{Res}_{P'}^{k(P')}(\varphi(z)\,\mathrm{d}\varphi(t)) = L'(\varphi(z)\partial_{\pi'}(\varphi(t)))_{-1} = L'(\varphi(z\partial_\pi(t)))_{-1} = (\overline{\varphi}_* L(z\partial_\pi(t)))_{-1} = \overline{\varphi}\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big).$$

Finally $\overline{\varphi}$ is an isomorphism of k-algebras, so $\operatorname{Tr}_{k(P')/k} \circ \overline{\varphi} = \operatorname{Tr}_{k(P)/k}$, and applying the trace gives the second identity. ∎

Proposition 6.3.45 (Elementary Properties of the Residue Symbol at a Place)
Let P be a place of K/k, π a uniformiser, $t \in K$. Then

a) $z \mapsto \operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)$ is k-linear $K \to k(P)$, and $z \mapsto \operatorname{Res}_P(z\,\mathrm{d}t)$ is k-linear $K \to k$;
b) for $x \in K$ transcendental over k and $\partial_x \in \operatorname{Der}_k(K, K)$ the distinguished derivation of (6.3.32), $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) = \operatorname{Res}_P^{k(P)}((z \cdot \partial_x(t))\,\mathrm{d}x)$ and $\operatorname{Res}_P(z\,\mathrm{d}t) = \operatorname{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x)$ for every $z \in K$;
c) if $z\partial_\pi(t) \in \mathcal{O}_P$ then $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) = 0$ and $\operatorname{Res}_P(z\,\mathrm{d}t) = 0$; in particular this holds if $z, t \in \mathcal{O}_P$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a) By (6.3.37), $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) = L_\pi(z\partial_\pi(t))_{-1}$; k-linearity in z follows from that of ∂_π, of L_π (6.3.35), and of coefficient extraction (6.3.36). The trace is k-linear.

b) By (6.3.32) (applied to $D := \partial_\pi$), $\partial_\pi = \partial_\pi(x) \cdot \partial_x$, so $z\partial_\pi(t) = (z \cdot \partial_x(t))\partial_\pi(x)$; applying $L_\pi(-)_{-1}$ to both sides (6.3.37) (once with t and once with x) gives the first identity, and applying $\operatorname{Tr}_{k(P)/k}$ the second.

c) If $z\partial_\pi(t) \in \mathcal{O}_P$ then $L_\pi(z\partial_\pi(t)) \in k(P)[[X]]$ (6.3.35), so its X^{-1}-coefficient vanishes, that is $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) = 0$, and so is its trace. If $z, t \in \mathcal{O}_P$ then $\partial_\pi(t) \in \mathcal{O}_P$ (6.3.42), so $z\partial_\pi(t) \in \mathcal{O}_P$. ∎

6.3.7 The Algebraic Residue Theorem over an Algebraically Closed Field

Assume throughout this subsection that k is algebraically closed. Then every place of K/k is rational (6.3.6).c), so the entire theory of (6.3.37) applies directly, with no need for the base-change-to-$\overline{k}$ machinery of the following subsections (which is needed only once k is no longer assumed algebraically closed).

Proposition 6.3.46 (Algebraic Residue Theorem, over an Algebraically Closed Field)
Assume k is algebraically closed. For $t, f \in K$,

$$\sum_P \operatorname{Res}_P(f\,\mathrm{d}t) = 0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Roadmap. The proof, given immediately below, proceeds in the following steps.

- *Reduce to a differential $f\,\mathrm{d}x$.* Choose a separating transcendental $x \in K$; then $f\,\mathrm{d}t = f\partial_x(t)\,\mathrm{d}x$ at every place (6.3.45).b).
- *The case $K = k(x)$.* Partial fractions over $k = \overline{k}$ (6.3.47) write a rational function as a sum of terms $c_{i,j}(x - \theta_i)^{-j}$; the finite residues are the $c_{i,1}$, and the place at infinity has residue $-\sum_i c_{i,1}$ (6.3.51).
- *Trace formula.* For a finite separable $K/k(x)$ and a place Q of $k(x)$, $\sum_{P|Q} \operatorname{Res}_P(f\,\mathrm{d}x) = \operatorname{Res}_Q\big(\operatorname{Tr}_{K/k(x)}(f)\,\mathrm{d}x\big)$ (6.3.54). This is proved locally, after completing at Q: the completed field splits as $\prod_i k((\pi_i))$ over the places

© David Rufino CC BY-NC-ND 4.0

above Q, reducing the identity to the local trace formula at each place, which is computed monomial by monomial via the roots of unity in k, following Lang.

- *Assemble.* Group the places P of K by the place Q of $k(x)$ below them. The trace formula turns each inner sum into $\mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$, and summing over Q gives 0 by the case $K = k(x)$.

Proof of (6.3.46). Drafted by Claude Sonnet, reviewed by David Rufino.

Fix any transcendental $x \in K$ with $K/k(x)$ finite and separable (exists as K/k is a function field, separably generated), and let $\partial_x \in \mathrm{Der}_k(K,K)$ be the distinguished element of (6.3.32) with $\partial_x(x) = 1$. By (6.3.45).b), for every place P,

$$\mathrm{Res}_P(f\,\mathrm{d}t) = \mathrm{Res}_P((f\cdot\partial_x(t))\,\mathrm{d}x)$$

this handles $t \in k$ automatically too, as $\partial_x(k) = 0$ then makes both sides 0, matching $\mathrm{d}t = 0$. So it suffices to prove $\sum_P \mathrm{Res}_P(f'\,\mathrm{d}x) = 0$ for $f' := f\cdot\partial_x(t) \in K$.

Every place P of K/k lies over a unique place Q of $k(x)/k$, so (finitely many terms being nonzero) the sum may be regrouped:

$$\sum_{P \text{ of } K/k} \mathrm{Res}_P(f'\,\mathrm{d}x) = \sum_{Q \text{ of } k(x)/k}\ \sum_{P\mid Q} \mathrm{Res}_P(f'\,\mathrm{d}x).$$

By (6.3.54), the inner sum is $\mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$ for every Q, so

$$\sum_P \mathrm{Res}_P(f'\,\mathrm{d}x) = \sum_Q \mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big) = 0$$

by (6.3.51), applied to $\mathrm{Tr}_{K/k(x)}(f') \in k(x)$. ∎

This proof uses only (6.3.32) (already available, with no need for the module of differentials $\Omega_{K/k}$ to have been formally constructed).

Lemma 6.3.47 (Partial Fraction Decomposition over an Algebraically Closed Field)
Let k be algebraically closed and $r, s \in k[X]$ polynomials with $s \neq 0$. Write $s(X) = c\prod_{i=1}^m (X-\theta_i)^{e_i}$ for distinct $\theta_i \in k$, integers $e_i \geq 1$, and $c \in k^\times$. Then there exist a polynomial $Q \in k[X]$ and constants $c_{i,j} \in k$ $(1 \leq i \leq m,\ 1 \leq j \leq e_i)$ with

$$\frac{r(X)}{s(X)} = Q(X) + \sum_{i=1}^m \sum_{j=1}^{e_i} \frac{c_{i,j}}{(X-\theta_i)^j}$$

as an identity in $k(X)$. If $\deg r < \deg s$ then $Q = 0$.

Proof. Absorbing c into r, we may take $c = 1$. We argue by induction on m, the number of distinct roots of s. If $m = 0$ then $s = 1$ and we take $Q := r$.

Let $m \geq 1$ and set $s_1 = (X-\theta_1)^{e_1}$ and $s_2(X) := \prod_{i=2}^m (X-\theta_i)^{e_i}$, so $s = s_1 s_2$. By (3.18.10) s_1 and s_2 are co-prime as they have no common roots in k, so $gs_1 + hs_2 = 1$ for some $g, h \in k[X]$ and

$$\frac{r(X)}{s(X)} = \frac{r(X)h(X)}{s_1}(X) + \frac{r(X)g(X)}{s_2}(X)$$

By the division algorithm $rh = Q_1 s_1 + r_1$ with $\deg(r_1) < e_1$. By (3.7.12) $r_1 = \sum_{l=0}^{e_1-1} d_l (X-\theta_1)^l$ for some $d_l \in k$, so with $c_{1,j} := d_{e_1-j}$

$$\frac{rh}{s_1} = Q_1 + \sum_{j=1}^{e_1} \frac{c_{1,j}}{(X-\theta_1)^j}$$

The induction hypothesis applies to rg/s_2, which has $m-1$ distinct roots in its denominator, and gives the terms with $i \geq 2$ together with a polynomial Q_2. Adding, the claim holds with $Q := Q_1 + Q_2$.

Finally suppose $\deg r < \deg s$. Multiplying the decomposition by s gives $Qs = r - \sum_{i,j} c_{i,j} s/(X-\theta_i)^j$, where each $s/(X-\theta_i)^j$ is a polynomial of degree $\deg s - j \leq \deg s - 1$. So the right hand side has degree less than $\deg s$, whereas $\deg(Qs) \geq \deg s$ if $Q \neq 0$; hence $Q = 0$. ∎

Remark 6.3.48 (A Complex-Analytic Heuristic)
Over $k = \mathbb{C}$, this also follows from Liouville's theorem. At each θ_i, choose $c_{i,1}, \ldots, c_{i,e_i}$ to match the principal part of the Laurent expansion of r/s at θ_i (possible and unique, exactly as the induction above extracts them); then $r/s -$

© David Rufino CC BY-NC-ND 4.0

$\sum_{i,j} c_{i,j}/(X-\theta_i)^j$ *is a rational function with no poles anywhere in* $\mathbb{C}$*, hence an entire function, and one that grows at most polynomially at infinity (being itself a rational function) – so by (the polynomial-growth form of) Liouville's theorem it is a polynomial, namely* Q*.*

Lemma 6.3.49 (Residue at Infinity of a Proper Rational Function)
Let $f, g \in k[X]$ *with* $g \neq 0 \neq f$ *and* $\deg(f) < \deg(g)$*. Then, for the place at infinity of* $k(X)/k$ *(6.3.10),*

$$\operatorname{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right) = \begin{cases} -\frac{\ell(f)}{\ell(g)} & \deg(f) = \deg(g) - 1 \\ 0 & \deg(f) < \deg(g) - 1. \end{cases}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $p := \deg(f)$, $q := \deg(g)$, and $m := q - p \geq 1$. Set $A(U) := U^p f(1/U)$, $B(U) := U^q g(1/U) \in k[X]$, polynomials with $A(0) = \ell(f)$, $B(0) = \ell(g) \neq 0$, so $w := A/B \in k[[U]]$ is a unit with $w(0) = \ell(f)/\ell(g)$. As $f(1/U) = U^{-p}A(U)$ and $g(1/U) = U^{-q}B(U)$,

$$\frac{f}{g}(1/U) = U^{q-p}\frac{A(U)}{B(U)} = U^m w(U).$$

Let U be an indeterminate and $\varphi : k(X) \to k(U)$ the isomorphism with $\varphi(X) = U^{-1}$; it maps the place at infinity to the place 0 of $k(U)$, with uniformiser U (6.3.10). By (6.3.44) and (6.3.45).b) (with $x := U$, as $\partial_U(U^{-1}) = -U^{-2}$),

$$\operatorname{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right) = \operatorname{Res}_0(\varphi(f/g)\,\mathrm{d}(U^{-1})) = \operatorname{Res}_0(-U^{m-2}w(U)\,\mathrm{d}U).$$

As $\partial_U(U) = 1$, this is the coefficient of U^{-1} in $-U^{m-2}w(U)$, that is − the coefficient of U^{1-m} in $w(U)$. If $m = 1$ this is $-w(0) = -\ell(f)/\ell(g)$. If $m \geq 2$ then $1 - m < 0$, and as $w \in k[[U]]$ has only non-negative powers, this coefficient is 0. ∎

Remark 6.3.50 (A Complex-Analytic Heuristic)
Over $k = \mathbb{C}$*, this is controlled entirely by the leading-order behaviour of* f/g *at infinity. Writing* $U = 1/X$*, the residue at infinity is* $-\operatorname{Res}_0$ *of the pulled-back differential, and* $f/g(1/U)\cdot\partial_U(U^{-1}) = -U^{m-2}w(U)$ *has leading term* $-\left(\frac{\ell(f)}{\ell(g)}\right)U^{m-2}$ *(*$w(0) = \ell(f)/\ell(g) \neq 0$*); classically* $\oint U^n\,\mathrm{d}U = 0$ *for every* $n \neq -1$*, so the residue vanishes unless this* leading *term is itself the* U^{-1} *term, i.e.* $m - 2 = -1$*, or* $\deg f = \deg g - 1$ *– exactly the borderline decay rate* $f/g \sim (\ell(f)/\ell(g))X^{-1}$ *at infinity – in which case the residue is read off directly from the leading coefficients, with no correction from the rest of the expansion of* w*.*

Lemma 6.3.51 (Residue Theorem for the Rational Function Field)
Assume k *is algebraically closed. For* $\theta \in k$*,* $n \in \mathbb{Z}$ *and* $g := (X-\theta)^n \in k(X)$*, the residues at the places of* $k(X)/k$ *(the* $\mathcal{O}_{X-\theta'}$*,* $\theta' \in k$*, written* θ'*, and the place at infinity (6.3.10)) are*

$$\operatorname{Res}_{\theta'}(g\,\mathrm{d}X) = \begin{cases} 1 & \theta' = \theta \text{ and } n = -1 \\ 0 & \text{otherwise} \end{cases} \qquad \operatorname{Res}_\infty(g\,\mathrm{d}X) = \begin{cases} -1 & n = -1 \\ 0 & n \neq -1. \end{cases}$$

Consequently, for every $g \in k(X)$*, where* $k(X)/k$ *is the rational function field,*

$$\sum_Q \operatorname{Res}_Q(g\,\mathrm{d}X) = 0,$$

summed over all places Q *of* $k(X)/k$ *(finitely many terms nonzero).*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write T for the variable of the Laurent series field, to avoid a clash with $X \in k(X)$, so that $L_\pi : k(X) \to k((T))$ is a ring homomorphism with $L_\pi(\pi) = T$ (6.3.35).

Finite places. The place θ' has uniformiser $\pi := X - \theta'$ with $\partial_\pi(X) = 1$. If $\theta' \neq \theta$ then g and X lie in $\mathcal{O}_{X-\theta'}$, so $\operatorname{Res}_{\theta'}(g\,\mathrm{d}X) = 0$ (6.3.45).c). If $\theta' = \theta$ then by (6.3.37)

$$\operatorname{Res}_\theta(g\,\mathrm{d}X) = L_\pi(\pi^n)_{-1} = (T^n)_{-1} = \begin{cases} 1 & n = -1 \\ 0 & n \neq -1 \end{cases}$$

The place at infinity. If $n \geq 0$ then g is a polynomial, so $g(1/U)$ is a k-linear combination of $U^0, \ldots, U^{-n}$, and $g(1/U)\partial_U(X) = -U^{-2}g(1/U)$ a linear combination of $U^{-2}, \ldots, U^{-n-2}$; as every exponent is ≤ -2, $\operatorname{Res}_\infty(g\,\mathrm{d}X) = 0$. If $n \leq -1$ then $g = 1/(X-\theta)^{-n}$ is a proper fraction with numerator 1 and denominator $(X-\theta)^{-n}$ of degree $-n$, so by (6.3.49)

© David Rufino CC BY-NC-ND 4.0

$$\operatorname{Res}_\infty(g\,\mathrm{d}X)=\begin{cases}-1 & n=-1\\ 0 & n\le -2.\end{cases}$$

The sum vanishes for every $g\in k(X)$. By (6.3.47) every $g\in k(X)$ is the sum of a polynomial and a k-linear combination of the functions $(X-\theta)^{-n}$ with $\theta\in k$, $n\ge 1$; and by (3.7.12) every polynomial is a k-linear combination of the $(X-\theta)^n$, $n\ge 0$, for any fixed $\theta\in k$. The residue being k-linear, this reduces to the case $g=(X-\theta)^n$ above, where the residues at all places other than θ and ∞ vanish, and $\operatorname{Res}_\theta(g\,\mathrm{d}X)+\operatorname{Res}_\infty(g\,\mathrm{d}X)$ is $1-1=0$ if $n=-1$ and 0 otherwise. ∎

Lemma 6.3.52 (Local Trace Formula for Residues)
Let k be algebraically closed, $F:=k((t))$, and let $E=k((\pi))$ be a finite separable extension of F with $v_\pi(t)=e$, so $[E:F]=e$ and $t=\pi^e w$ for some $w\in k[[\pi]]^\star$ (3.32.35). For $g\in E$ write $\operatorname{Res}_\pi(g\,\mathrm{d}t):=(g\partial_\pi(t))_{-1}$. Then, in every characteristic,

$$\operatorname{Res}_t\bigl(\operatorname{Tr}_{E/F}(g)\,\mathrm{d}t\bigr)=\operatorname{Res}_\pi(g\,\mathrm{d}t)\qquad(g\in E).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Basis and matrices. By (3.32.31) (with $R:=k$, $X:=\pi$, $m:=e$, $\tau=t$), every element of $k[[\pi]]$ is uniquely $\sum_{i<e}f_i(t)\pi^i$ with $f_i\in k[[Y]]$; multiplying by powers of t (which is π^e up to a unit) gives the same for E. So $1,\pi,\dots,\pi^{e-1}$ is an F-basis of E. For $g\in E$ write $g\pi^i=\sum_{j<e}M_{ji}(g)\pi^j$ with $M_{ji}(g)\in F$, and $\operatorname{Tr}_{E/F}(g)=\sum_{i<e}M_{ii}(g)$.

Universal model. Let $R:=\mathbb{Z}\bigl[b_0^{\pm1},b_1,b_2,\dots\bigr]$, a domain of characteristic 0, and put $w_R:=\sum_{i\ge0}b_i\pi^i\in R[[\pi]]^\star$, $t_R:=\pi^e w_R$. The construction above works verbatim over R (only b_0 needs inverting), with matrix entries $M_{ji}(g)\in R((t_R))$ given by universal formulas in $+$, $\times$ and w_R^{-1}, whose coefficients lie in R. For $r\in\mathbb{Z}$ set

$$\Psi(\pi^r):=\operatorname{Res}_{t_R}(\operatorname{Tr}_R(\pi^r)\,\mathrm{d}t_R)-\operatorname{Res}_\pi(\pi^r\,\mathrm{d}t_R)\in R,$$

both terms being coefficients of elements of $R((\pi))$ or $R((t_R))$.

Claim: $\Psi(\pi^r)=0$ for all r. Let Ω be an algebraic closure of $\operatorname{Frac}(R)$; it has characteristic 0, so e is invertible in Ω and $R\subset\Omega$. Write $w_R=b_0u$ with $u\in1+\pi R[[\pi]]$. The binomial series $\varepsilon:=\sum_{i\ge0}\binom{1/e}{i}(u-1)^i\in\Omega[[\pi]]$ satisfies $\varepsilon^e=u$, and for $\beta\in\Omega$ with $\beta^e=b_0$ the element $\varpi:=\beta\pi\varepsilon$ satisfies $\varpi^e=t_R$. As $\varpi=\pi\cdot(\text{unit})$, it is a uniformiser of $\Omega((\pi))=\Omega((\varpi))$, and $\Omega((t_R))\subset\Omega((\pi))$ has basis $1,\varpi,\dots,\varpi^{e-1}$ (the argument above, with ϖ in place of π).

Take $g=\varpi^r$ with $r=es+j$, $0\le j<e$. Then $\varpi^r=t_R^s\varpi^j$, and multiplication by ϖ^j sends the basis vector ϖ^i to ϖ^{i+j}, which is a basis vector times t_R^0 or t_R. So the matrix of ϖ^r has nonzero entries only in positions shifted by r modulo e: its diagonal vanishes unless $e\mid r$, and if $r=es$ it is the scalar t_R^s. Hence

$$\operatorname{Tr}(\varpi^r)=\begin{cases}et_R^s & r=es\\ 0 & e\nmid r.\end{cases}$$

Therefore $\operatorname{Res}_{t_R}(\operatorname{Tr}(\varpi^r)\,\mathrm{d}t_R)=e$ if $r=-e$ and 0 otherwise. On the other side $\mathrm{d}t_R=e\varpi^{e-1}\,\mathrm{d}\varpi$, so $\varpi^r\,\mathrm{d}t_R=e\varpi^{r+e-1}\,\mathrm{d}\varpi$ and $\operatorname{Res}_\varpi(\varpi^r\,\mathrm{d}t_R)$ is again e for $r=-e$ and 0 otherwise. Residues are independent of the choice of uniformiser (6.3.43), so $\operatorname{Res}_\varpi(g\,\mathrm{d}t_R)=\operatorname{Res}_\pi(g\,\mathrm{d}t_R)$. Thus the identity holds for $g=\varpi^r$ for every r. Both sides are Ω-linear and continuous in g, and the ϖ^r span $\Omega((\pi))$ topologically, so it holds for every $g\in\Omega((\pi))$; in particular $\Psi(\pi^r)\in R$ maps to 0 in Ω. As $R\subset\Omega$, $\Psi(\pi^r)=0$.

Specialisation. Given $E=k((\pi))$ and $t=\pi^e w$ with $w=\sum_{i\ge0}c_i\pi^i$, $c_0\in k^\times$, let $\phi:R\to k$ be the ring homomorphism with $b_i\mapsto c_i$ (it exists by the universal property of localisation, as c_0 is a unit). Applying ϕ coefficientwise gives ring homomorphisms $R((\pi))\to k((\pi))$ and $R((t_R))\to k((t))$ (with $t_R\mapsto t$, since $\phi(w_R)=w$). These commute with the formation of matrices M_{ji}, hence with traces, and with the coefficient extraction defining both residues. So $\phi(\Psi(\pi^r))$ is the defect $\operatorname{Res}_t\bigl(\operatorname{Tr}_{E/F}(\pi^r)\,\mathrm{d}t\bigr)-\operatorname{Res}_\pi(\pi^r\,\mathrm{d}t)$ over k, and it equals 0. Finally, for $g=\sum_r a_r\pi^r\in k((\pi))$ both sides depend on only finitely many a_r, since $\operatorname{Tr}(\pi^r)$ has t-order tending to ∞ with r and only finitely many terms contribute to a given coefficient of t^{-1} or π^{-1}. The identity for each monomial therefore gives it for g. ∎

Lemma 6.3.53 (Places over P and Irreducible Factors)
Let k be algebraically closed, $K/k(x)$ a finite separable extension of degree n, P the place (x) of $k(x)$, $K_P:=k((x))$, and $y\in K$ a primitive element with minimal polynomial g over $k(x)$. Factor $g=g_1\cdots g_r$ into distinct monic irreducibles over K_P, with $d_j:=\deg g_j$, and let $E_j:=K_P[Y]/(g_j)$ with $\iota_j:K\hookrightarrow E_j$ sending $y\mapsto Y$. Then $j\mapsto Q_j$, defined below, is a bijection from $\{1,\dots,r\}$ onto the places of K/k over P, and $e_{Q_j}=d_j$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

© David Rufino CC BY-NC-ND 4.0

The place Q_j. E_j/K_P is finite separable (as g_j is separable), so by (3.32.35) there is an isomorphism $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ over K_P with $\Psi_j(x)$ of order d_j. Pulling back the valuation of $k\big((\pi_j)\big)$ along $\Psi_j \circ \iota_j$ gives a discrete valuation of K extending v_P with ramification index d_j; let Q_j be the corresponding place.

Every place is some Q_j. Let Q be a place of K/k over P. As k is algebraically closed, Q is rational, so (6.3.35) gives a valuation-preserving embedding $L_Q : K \hookrightarrow k\big((\pi_Q)\big)$. Since $g(y) = 0$, the element $L_Q(y)$ is a root of g in $k\big((\pi_Q)\big)$. The g_j are pairwise coprime in $K_P[Y]$, so $L_Q(y)$ is a root of exactly one g_j (a common root of $g_j, g_{j'}$ would be a root of 1, by Bézout). Call this $j = j(Q)$. Then L_Q factors through E_j, and $L_Q(x)$ has ramification index e_Q over K_P; the valuation induced on E_j extends v_P, so by the uniqueness of extension (3.32.36) (applied to E_j/K_P) it equals the valuation of Q_j. Hence $Q = Q_{j(Q)}$.

Injectivity. If $Q_j = Q_{j'}$ then $j(Q_j) = j$ and $j(Q_{j'}) = j'$ are determined by the same valuation, and the minimal factor of $L(y)$ is unique, so $j = j'$. ∎

Proposition 6.3.54 (Trace Formula for Residues)
Assume k is algebraically closed. Let $\mathcal{O} := k[x]$, $\mathcal{B} := \overline{\mathcal{O}}^K$ the integral closure of $\mathcal{O}$ in a finite separable extension $K/k(x)$, P a place of $k(x)/k$ corresponding to a maximal ideal $\mathfrak{p} \lhd \mathcal{O}$, and $f \in K$. Then

$$\sum_{Q|P} \mathrm{Res}_Q(f\,\mathrm{d}x) = \mathrm{Res}_P\Big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\Big),$$

with no hypothesis on $\mathrm{char}(k)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As k is algebraically closed every place is rational, so $\mathfrak{p} = (x - \theta)$ for some $\theta \in k$; replacing x by $x - \theta$ we may assume $\mathfrak{p} = (x)$.

By (3.31.19) the places $Q_1, \dots, Q_r$ of K/k over P have ramification indices $e_1, \dots, e_r$ with $\sum_i e_i = n := [K : k(x)]$ (residue degree 1, as k is algebraically closed). Let $K_P := k((x))$, with $k(x) \hookrightarrow K_P$ the expansion at P (6.3.35).

Factoring the minimal polynomial. Let $y \in \mathcal{B}$ be a primitive element with minimal polynomial $g \in \mathcal{O}[Y]$, monic of degree n. Factor $g = g_1 \cdots g_r$ over K_P as in (6.3.53), with $E_j = K_P[Y]/(g_j)$, $\iota_j : K \hookrightarrow E_j$, and $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ the isomorphisms given there, so that $\Psi_j(x)$ has order $d_j = \deg g_j$. Put $\lambda_j := \Psi_j \circ \iota_j : K \hookrightarrow k\big((\pi_j)\big)$ and $L_{\pi_j} := \lambda_j$. By (6.3.53), the places Q_j attached to the g_j are exactly the places of K/k over P, with $e_{Q_j} = d_j$, so after relabelling Q_i is the place attached to g_i and $e_i = d_i$.

Trace additivity. By the Chinese remainder theorem applied to $g = \prod_j g_j$,

$$K \otimes_{k(x)} K_P \xrightarrow{\sim} \prod_{i=1}^{r} E_i, \quad f \otimes 1 \mapsto (\iota_i(f))_i,$$

and composing with the isomorphisms Ψ_i identifies the right side with $\prod_i k((\pi_i))$, with $f \otimes 1 \mapsto \big(L_{\pi_i}(f)\big)_i$. The trace is unchanged by base change and additive over a finite product, so

$$\mathrm{Tr}_{K/k(x)}(f) = \sum_{i=1}^{r} \mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big) \qquad (f \in K).$$

Local identities. The residue at Q_i is the residue symbol of (6.3.37), computed in $k((\pi_i))$ through L_{π_i}, so $\mathrm{Res}_{Q_i}(f\,\mathrm{d}x) = \mathrm{Res}_{\pi_i}\big(L_{\pi_i}(f)\,\mathrm{d}x\big)$ (Laurent residue of (6.3.36)), and $\mathrm{Res}_P(z\,\mathrm{d}x) = \mathrm{Res}_x(z\,\mathrm{d}x)$ for $z \in K_P = k((x))$. Applying (6.3.52) to $E = k((\pi_i))$ over $F = K_P$ (with $t = x$ and $\Psi_i(x) = \tau_i$) gives, for every $f \in K$,

$$\mathrm{Res}_{Q_i}(f\,\mathrm{d}x) = \mathrm{Res}_P\Big(\mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\Big).$$

Assembly. Summing over i and using k-linearity of Res_P and trace additivity,

$$\sum_{Q|P} \mathrm{Res}_Q(f\,\mathrm{d}x) = \sum_{i=1}^{r} \mathrm{Res}_P\Big(\mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\Big) = \mathrm{Res}_P\Big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\Big).$$

∎

6.3.8 Base Extension of the Constant Field

Throughout this section, in addition to the standing hypothesis that k is the full constant field of K ($\tilde{k} = k$), assume also that k is **perfect**, so that K/k is a regular extension (3.35.45). Let k'/k be an algebraic field extension, and write $K_{k'} := K \otimes_k k'$.

© David Rufino CC BY-NC-ND 4.0

Proposition 6.3.55 (The Base-Changed Function Field)
$K_{k'}$ is a field, and an algebraic function field over k', which is a regular extension; if k'/k is finite, $[K_{k'} : K] = [k' : k]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$K_{k'}$ is a field. As K/k is a regular extension and k'/k is algebraic, $K_{k'} = K \otimes_k k'$ is a field (3.35.45).

$K_{k'}/k'$ is a regular extension. By associativity of tensor products (3.4.13) and $\overline{k'} = \overline{k}$ (k'/k algebraic), $K_{k'} \otimes_{k'} \overline{k'} \cong K \otimes_k \left(k' \otimes_{k'} \overline{k}\right) = K \otimes_k \overline{k}$, a field, as K/k is a regular extension (3.35.45).

$K_{k'}/k'$ is an algebraic function field. Fix a transcendental $x \in K$ with $K/k(x)$ finite (exists, as K/k is an algebraic function field). As $k[x] \otimes_k k' \cong k'[x]$ (tensoring a polynomial ring with a field extension), the image of x in $K_{k'}$ remains transcendental over k'. The image of $k(x) \otimes_k k'$ in $K_{k'}$ (well-defined and injective, as $k(x) \subset K$ is a k-subalgebra, (3.35.2)) is then a subring of the field $K_{k'}$, hence a domain; as k'/k is algebraic, it is thus a field (3.34.57), and – being the subfield of $K_{k'}$ generated by k' and the transcendental x – equals $k'(x)$. By associativity of tensor products (3.4.13) (as K is a $k(x)$-module),

$$K_{k'} = K \otimes_k k' \cong K \otimes_{k(x)} (k(x) \otimes_k k') = K \otimes_{k(x)} k'(x)$$

As $K/k(x)$ is finite, K is a finite-dimensional $k(x)$-vector space, hence free (3.3.126) with a basis of size $n := [K : k(x)]$; extension of scalars (3.4.26) then makes $K_{k'} \cong K \otimes_{k(x)} k'(x)$ a free $k'(x)$-module with a basis of the same size n, in particular finite over $k'(x)$. So $K_{k'}$ is a finitely generated field extension of k' (generated by x together with a finite $k'(x)$-basis) of transcendence degree 1: an algebraic function field.

Degree. If k'/k is finite, k' is a free k-module with a basis of size $[k' : k]$, so extension of scalars along $k \hookrightarrow K$ (3.4.26) makes $K_{k'} = K \otimes_k k'$ a free K-module with a basis of the same size, giving $[K_{k'} : K] = [k' : k]$. ∎

Proposition 6.3.56 (Constant Field Extension is Unramified)
Let P be a place of K/k and k'/k algebraic. Every place P' of $K_{k'}/k'$ lying over P is unramified.

There are exactly r such places, where r is the number of maximal ideals of $A := k(\mathfrak{m}_P) \otimes_k k'$ (finite, even when k'/k is not). Moreover each such P' satisfies $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'} \cap K = \mathfrak{m}_P$.

More precisely, writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, $A \cong \mathcal{O}_{P,k'}/\mathfrak{m}_P\mathcal{O}_{P,k'}$ as k'-algebras (3.4.42) there is a bijection

$$\begin{aligned} \operatorname{Specm}(k(\mathfrak{m}_P) \otimes_k k') &\longleftrightarrow \{\text{places of } K_{k'}/k' \text{ lying over } P\} \\ \mathfrak{p}'/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longrightarrow \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}'} \\ \left(\mathfrak{m}_{P'} \cap \mathcal{O}_{P,k'}\right)/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longleftarrow \mathcal{O}_{P'} \end{aligned}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We claim that we are in the situation of (3.35.47), applied with $\mathcal{O} := \mathcal{O}_P$, $F := K$:

- $\mathcal{O}_P$ is a discrete valuation ring containing k (a place of K/k is by definition a valuation ring $k \subsetneq \mathcal{O}_P \subsetneq K$ (6.3.3), automatically discrete (6.3.7))
- with fraction field K whose full constant field is k (the standing hypothesis of this section)
- $k(\mathfrak{m}_P)$ is finite over k (6.3.6).c)
- every valuation ring R of $K_{k'}/k'$ with $k' \subset R$ and $\mathcal{O}_P \subset R$ satisfies $R \cap K = \mathcal{O}_P$: such an R is exactly a place of $K_{k'}/k'$, and this is (6.3.12) (applied with $L := K_{k'}$, $k'' := k'$)

Therefore the result follows from there: writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, its $\mathfrak{b} := \mathfrak{m}_P\mathcal{O}_{P,k'}$ has quotient $\mathcal{O}_{P,k'}/\mathfrak{b} \cong A$ (above), so a) gives exactly r maximal ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of $\mathcal{O}_{P,k'}$; by b), each $\left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ is a place P'_j of $K_{k'}/k'$ (with $\mathcal{O}_{P'_j} := \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$), unramified over $\mathcal{O}_P$, and, as in the statement, corresponds to $\mathfrak{p}_j/\mathfrak{b} \in \operatorname{Specm}(A)$; and, by c) together with the last point above, *every* place P' of $K_{k'}/k'$ over P has $\mathcal{O}_{P'} = \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ for some j, that is $P' = P'_j$, giving the stated bijection. As $K_{k'}/k'$ is itself an algebraic function field (6.3.55), every place of $K_{k'}/k'$ – in particular P'_j – is automatically a discrete valuation ring (6.3.7), so $e\left(P'_j/P\right) = v_{P'_j}(\pi_P) = 1$: unramified, with π_P a uniformiser of $\mathcal{O}_P$ (3.30.2). Finally $\mathcal{O}_{P'_j} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'_j} \cap K = \mathfrak{m}_P$ by e). ∎

Lemma 6.3.57 (A Rational Place Extends Uniquely Under Any Algebraic Base Change)
Let P be a rational place of K/k. Then for any algebraic k'/k, there is a unique place Q of $K_{k'}$ over P, and $e(Q/P) = 1$, with Q again rational over k'.

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is the case $k(\mathfrak{m}_P) = k$ of (6.3.56). ∎

Proposition 6.3.58 (Complete Splitting)
Let K/k be an algebraic function field, P a place of K/k, and k'/k an algebraic extension. Suppose further that k' contains a normal closure of $k(\mathfrak{m}_P)$.

*Then P **splits completely** in $K_{k'}/k'$: that is there are exactly $n = [k(\mathfrak{m}_P) : k]$ places above P, each rational over k'.*

This holds in particular when $k' = \overline{k}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $k_P := k(\mathfrak{m}_P)$; k_P/k is finite (6.3.6).c) and separable, as k is perfect (3.18.103). As k' contains a normal closure of k_P/k, (3.35.15) gives $k_P \otimes_k k' \cong (k')^n$ as k'-algebras, with exactly n maximal ideals, each with residue field k' (the coordinate kernels). By (6.3.56) (with $A := k_P \otimes_k k'$) this implies that P splits completely into n places, each rational (residue field k'). ∎

Proposition 6.3.59 (Automorphisms Act Transitively on Places over $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P (6.3.58) (with $k' := \overline{k}$). For $\sigma \in \mathrm{Aut}\left(\overline{k}/k\right)$, the map

$$\sigma_* : K_{\overline{k}} \to K_{\overline{k}}, \qquad x \otimes c \mapsto x \otimes \sigma(c)$$

is a well-defined ring automorphism of $K_{\overline{k}}$ fixing K pointwise, and $\sigma \mapsto \sigma_*$ is a group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $K_{\overline{k}}$.

For any place P' of $K_{\overline{k}}/\overline{k}$ lying over P then $\sigma_*(\mathcal{O}_{P'})$ is a place lying over P and

$$v_{\sigma(P')}(\sigma_*(z)) = v_{P'}(z)$$

for all $z \in K_{\overline{k}}^\star$; this defines a transitive group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $\{P'_1, ..., P'_d\}$.

Moreover, for P' among $P'_1, ..., P'_d$ (rational over $\overline{k}$), write $\iota_{P'} : k(\mathfrak{m}_P) \hookrightarrow k(\mathfrak{m}_{P'}) = \overline{k}$ for the residue embedding induced by $\mathcal{O}_P \subset \mathcal{O}_{P'}$ (well-defined as $\mathfrak{m}_P \subset \mathfrak{m}_{P'} \cap \mathcal{O}_P$). Then $\iota_{\sigma(P')} = \sigma \circ \iota_{P'}$.

This yields a bijection

$$\begin{aligned} \{\text{places of } \ K/k\} &\longleftrightarrow \left\{\mathrm{Aut}\left(\overline{k}/k\right)\text{-orbits of places of } \ K_{\overline{k}}/\overline{k}\right\} \\ P &\longrightarrow \{P'_1, ..., P'_d\} \end{aligned}$$

Proof. As $\sigma : \overline{k} \to \overline{k}$ is a k-algebra homomorphism, $1_K \otimes \sigma$ is a well-defined k-algebra endomorphism of $K \otimes_k \overline{k} = K_{\overline{k}}$ (3.4.36).

As $(1_K \otimes \sigma) \circ (1_K \otimes \tau) = 1_K \otimes (\sigma\tau)$ then σ_* is an automorphism of $K_{\overline{k}}/K$ and $\sigma \mapsto \sigma_*$ is a group homomorphism.

As σ_* is a field automorphism of $K_{\overline{k}}$ then $\sigma_*(\mathcal{O}_{P'})$ is a place of $K_{\overline{k}}$ which lies over the same place as $\mathcal{O}_{P'}$, as σ_* fixes $K \supset \mathcal{O}_P$.

The valuation identity follows because σ_* restricts to an isomorphism on the valuation rings.

As k is perfect, $k(\mathfrak{m}_P)/k$ is separable (3.18.103), so (3.35.15) gives a canonical isomorphism of $\overline{k}$-algebras

$$k(\mathfrak{m}_P) \otimes_k \overline{k} \cong \overline{k}^d, \quad p(\theta) \otimes c \mapsto (c\,p(\theta_1), ..., c\,p(\theta_d)) \qquad \left(p \in k[X], c \in \overline{k}\right)$$

identifying the d maximal ideals of this fibre with $\mathfrak{p}_j$, the kernel of the j-th coordinate.

Let $\mathfrak{p}'_1, ..., \mathfrak{p}'_d$ denote the corresponding maximal ideals of $\mathcal{O}_{P,\overline{k}} = \mathcal{O}_P \otimes_k \overline{k}$. Define $\mathcal{O}_{P'_i} := \left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_i}$ then by (6.3.56) these are precisely the places $P'_1, ..., P'_d$ above P, and $\mathcal{O}_{P'_i} \cap K = \mathcal{O}_P$, which shows that the given map is injective.

By (3.18.66) $\mathrm{Aut}\left(\overline{k}/k\right)$ permutes the roots θ_i transitively. We may show that if $\sigma(\theta_j) = \theta_{j'}$ then $\sigma_*\left(\mathcal{O}_{P'_j}\right) = \mathcal{O}_{P'_{j'}}$ and the action on places is transitive and $\{P'_1, ..., P'_d\}$ is precisely an $\mathrm{Aut}\left(\overline{k}/k\right)$-orbit.

Given a place $\overline{P}$ of $K_{\overline{k}}/\overline{k}$, $\mathcal{O} := \mathcal{O}_{\overline{P}} \cap K$ is a (proper) valuation ring of K by (6.3.11), giving a place P of K/k with $\mathcal{O}_P = \mathcal{O} \subset \mathcal{O}_{\overline{P}}$, that is $\overline{P}$ lies over P. As $P'_1, ..., P'_d$ are all the places of $K_{\overline{k}}/\overline{k}$ above P (above), $\overline{P}$ is one of them. So the given map is a surjection. ∎

Lemma 6.3.60 (Residue Embeddings are Exactly the Embeddings into $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P (6.3.58) (with $k' := \overline{k}$).

© David Rufino CC BY-NC-ND 4.0

Denote by $\iota_j : k(\mathfrak{m}_P) \hookrightarrow k\left(\mathfrak{m}_{P'_j}\right) = \overline{k}$ *the corresponding embeddings, then these exhaust all the* k*-algebra embeddings* $k(\mathfrak{m}_P)/k \hookrightarrow \overline{k}/k$.

Proof. By (6.3.59) there are $d = [k(\mathfrak{m}_P) : k]$ places and, as k is perfect so $k(\mathfrak{m}_P)/k$ is separable (3.18.103), there are $\#\operatorname{Mor}_k\left(k(\mathfrak{m}_P), \overline{k}\right) = [k(\mathfrak{m}_P) : k]_s = d$ k-algebra embeddings (3.18.83).

We show that $j \mapsto \iota_j$ is injective, i.e. that $\iota_j = \iota_{j'}$ implies $\mathcal{O}_{P'_j} = \mathcal{O}_{P'_{j'}}$ (and so $j = j'$). Consider the composite $\rho_j : \mathcal{O}_{P,\overline{k}} \to \left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_j} = \mathcal{O}_{P'_j} \to \mathcal{O}_{P'_j}/\mathfrak{m}_{P'_j} = \overline{k}$. It is $\overline{k}$-linear and is uniquely determined by

$$\rho_j(a \otimes c) = c\,\iota_j(\overline{a}) \qquad \left(a \in \mathcal{O}_P, c \in \overline{k}\right),$$

so $\iota_j = \iota_{j'} \Longrightarrow \rho_j = \rho_{j'}$, and their kernels agree, that is $\mathfrak{p}'_j = \mathfrak{p}'_{j'}$ (as $\ker \rho_j = \mathfrak{m}_{P'_j} \cap \mathcal{O}_{P,\overline{k}} = \left(\mathfrak{p}'_j\left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_j}\right)^c = \mathfrak{p}'_j$ (3.5.42)). By (6.3.56) $P'_j = P'_{j'}$ and we are done by counting. ∎

6.3.9 Residues and Base Change

We compare the residue symbol at a place P of K/k with the residues at the rational places of $K_{\overline{k}}/\overline{k}$ above P. The place P splits completely into $d := [k(P) : k]$ *rational* places of $K_{\overline{k}}/\overline{k}$, each carrying its own $\overline{k}$-valued Laurent coefficient theory, and these residues are the images $\iota_j\left(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\right)$ of the single element $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) \in k(P)$ under the d embeddings of $k(P)$ in $\overline{k}$.

Definition 6.3.61 (Conjugate Places)
Let P *be a place of* K/k, $d := [k(\mathfrak{m}_P) : k]$, *and* $\iota_1, \ldots, \iota_d : k(\mathfrak{m}_P) \hookrightarrow \overline{k}$ *the embeddings, with* P'_j *the corresponding place of* $K_{\overline{k}}/\overline{k}$ *over* P, $\iota_j = \iota_{P'_j}$ *(6.3.60). As each* P'_j/P *is unramified, every uniformiser* π *at* P *remains a uniformiser at* P'_j *(6.3.56), so the residue symbol* $\operatorname{Res}_{P'_j}(w\,\mathrm{d}s)$ *at the rational place* P'_j *of* $K_{\overline{k}}/\overline{k}$ *(6.3.37) is defined for* $w, s \in K_{\overline{k}}$.

Lemma 6.3.62 (Laurent Coefficients and Base Change)
Let P *be a place of* K/k, π *a uniformiser, and* P'_j, ι_j $(1 \le j \le d)$ *as in (6.3.61). Write* $L_j : K_{\overline{k}} \hookrightarrow \overline{k}((X))$ *for the Laurent coefficient embedding at the rational place* P'_j *relative to the uniformiser* $\pi \otimes 1$, *and* $(\iota_j)_*$ *for the map* $k(P)((X)) \to \overline{k}((X))$ *applying* ι_j *to the coefficients. Then for every* $z \in K$

$$L_j(z \otimes 1) = (\iota_j)_* L_\pi(z).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The map $\Phi := (\iota_j)_* \circ L_\pi : K \to \overline{k}((X))$ is a k-algebra homomorphism, so $\tilde{\Phi}(z \otimes c) := c\,\Phi(z)$ defines an $\overline{k}$-algebra homomorphism $\tilde{\Phi} : K_{\overline{k}} = K \otimes_k \overline{k} \to \overline{k}((X))$. Recall $\mathcal{O}_{P'_j} = \left(\mathcal{O}_P \otimes_k \overline{k}\right)_{\mathfrak{p}'_j}$ with $\mathfrak{p}'_j = \ker \rho_j$ for $\rho_j(a \otimes c) = c\,\iota_j(\overline{a})$ (6.3.60). For $a \in \mathcal{O}_P$, $c \in \overline{k}$ the constant term of $\tilde{\Phi}(a \otimes c) = c\,(\iota_j)_* L_\pi(a)$ is $c\,\iota_j(\overline{a}) = \rho_j(a \otimes c)$, by (6.3.35) (the identity on residue fields). So $\tilde{\Phi}$ maps $\mathcal{O}_P \otimes_k \overline{k}$ into $\overline{k}[[X]]$, an element of $\mathfrak{p}'_j$ to $X\overline{k}[[X]]$ and an element outside $\mathfrak{p}'_j$ to a unit. Hence $\tilde{\Phi}$ extends to a local homomorphism $\mathcal{O}_{P'_j} \to \overline{k}[[X]]$ sending the uniformiser $\pi \otimes 1$ to X, and therefore $v_X\left(\tilde{\Phi}(z')\right) = v_{P'_j}(z')$ for $z' \in K_{\overline{k}}^\star$ (write $z' = (\pi \otimes 1)^n u$ with u a unit of $\mathcal{O}_{P'_j}$), and its constant term on $\mathcal{O}_{P'_j}$ is the residue class in $\overline{k} = k\left(P'_j\right)$. The uniqueness in (6.3.35), applied to the place P'_j of the function field $K_{\overline{k}}/\overline{k}$ (6.3.55) and the uniformiser $\pi \otimes 1$, gives $\tilde{\Phi} = L_j$. Restricting to $K \otimes 1$ gives the claim. ∎

Proposition 6.3.63 (Conjugate Residues)
Let P *be a place of* K/k *and* $t, z \in K$. *With the notation of (6.3.61), for every* $j = 1, \ldots, d$

$$\operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \iota_j\left(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\right),$$

and consequently

$$\sum_{j=1}^{d} \operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \operatorname{Tr}_{k(P)/k}\left(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\right) = \operatorname{Res}_P(z\,\mathrm{d}t) \in k.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix a uniformiser π at P; then $\pi \otimes 1$ is a uniformiser at P'_j (6.3.61). Applying the ring homomorphism $(\iota_j)_*$, which commutes with products, formal derivatives and coefficient extraction, and using $L_j(z \otimes 1) = (\iota_j)_* L_\pi(z)$ for z and for t (6.3.62),

$$\operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \operatorname{Res}_X\left(L_j(z \otimes 1)\,\mathrm{d}L_j(t \otimes 1)\right) = \operatorname{Res}_X\left((\iota_j)_* L_\pi(z)\,\mathrm{d}(\iota_j)_* L_\pi(t)\right) = \iota_j(\operatorname{Res}_X(L_\pi(z)\,\mathrm{d}L_\pi(t))) = \iota_j\left(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\right).$$

© David Rufino CC BY-NC-ND 4.0

By (6.3.60), $\mathrm{Mor}_k\big(k(P),\overline{k}\big)=\{\iota_1,\dots,\iota_d\}$; as k is perfect, $k(P)/k$ is separable (3.18.103), so $[k(P):k]_i=1$ (3.18.93). So (3.18.129) (trace is the sum of the embeddings in $\overline{k}$) gives, with $\alpha:=\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)$,

$$\mathrm{Tr}_{k(P)/k}(\alpha)=\sum_{j=1}^{d}\iota_j(\alpha)=\sum_{j=1}^{d}\mathrm{Res}_{P_j'}((z\otimes 1)\,\mathrm{d}(t\otimes 1)).$$

■

Corollary 6.3.64 (Residue Symbol as a Trace of the Residue over the Algebraic Closure)
Let P be a place of K/k and $t,z\in K$, with $P_1',\dots,P_d'$ the conjugate rational places of $K_{\overline{k}}/\overline{k}$ over P and $\iota_j:k(P)\hookrightarrow\overline{k}$ the corresponding embeddings (6.3.61). Then the residue over k is the trace of the residue over $\overline{k}$: for every j the residue $\mathrm{Res}_{P_j'}((z\otimes 1)\,\mathrm{d}(t\otimes 1))$ lies in $\iota_j(k(P))$ and

$$\mathrm{Res}_P(z\,\mathrm{d}t)=\mathrm{Tr}_{\iota_j(k(P))/k}\Big(\mathrm{Res}_{P_j'}((z\otimes 1)\,\mathrm{d}(t\otimes 1))\Big).$$

Proof. Write $\alpha:=\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)\in k(P)$. By (6.3.63), $\mathrm{Res}_{P_j'}((z\otimes 1)\,\mathrm{d}(t\otimes 1))=\iota_j(\alpha)\in\iota_j(k(P))$, and ι_j is an isomorphism $k(P)\to\iota_j(k(P))$ over k, so $\mathrm{Tr}_{\iota_j(k(P))/k}\big(\iota_j(\alpha)\big)=\mathrm{Tr}_{k(P)/k}(\alpha)=\mathrm{Res}_P(z\,\mathrm{d}t)$. ■

Once the module of differentials $\Omega_{K/k}$ is available, it is a 1-dimensional K-vector space dual to $\mathrm{Der}_k(K,K)$ (both being $\mathrm{trdeg}_k(K)=1$-dimensional, (3.34.44)), and the pairing of $z\,\mathrm{d}t\in\Omega_{K/k}$ with ∂_π is exactly $z\,\partial_\pi(t)$; $\mathrm{Res}_P(z\,\mathrm{d}t)$ is then the residue, in this sense, of the differential $z\,\mathrm{d}t$.

6.3.10 The Algebraic Residue Theorem

This subsection proves the (algebraic) Residue Theorem for a general base field k, by reduction to the algebraically closed case of (6.3.46).

Proposition 6.3.65 (Algebraic Residue Theorem)
For $t,f\in K$,

$$\sum_P\mathrm{Res}_P(f\,\mathrm{d}t)=0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (6.3.59) (applied to K/k), restriction $\overline{P}\mapsto\overline{P}\cap K$ is a bijection $\{\text{places of } K/k\}\leftrightarrow\Big\{\mathrm{Aut}\big(\overline{k}/k\big)\text{-orbits of places of } K_{\overline{k}}/\overline{k}\Big\}$ (where $K_{\overline{k}}:=K\otimes_k\overline{k}$, a field by (3.35.45) as in (6.3.55)); the fibre over P is $\big\{P_1',\dots,P_{d_P}'\big\}$, $d_P:=\deg P$. By (6.3.63), for every place P of K/k and $z\in K$,

$$\mathrm{Res}_P(z\,\mathrm{d}t)=\sum_{j=1}^{d_P}\mathrm{Res}_{P_j'}((z\otimes 1)\,\mathrm{d}(t\otimes 1)).$$

Summing over all places P of K/k and using that the fibres $\big\{P_1',\dots,P_{d_P}'\big\}$ partition the places of $K_{\overline{k}}/\overline{k}$ (restriction being a surjection),

$$\sum_{P\text{ of }K/k}\mathrm{Res}_P(f\,\mathrm{d}t)=\sum_{\overline{P}\text{ of }K_{\overline{k}}/\overline{k}}\mathrm{Res}_{\overline{P}}((f\otimes 1)\,\mathrm{d}(t\otimes 1))=0$$

by (6.3.46) (applicable as $K_{\overline{k}}/\overline{k}$ is itself an algebraic function field, over the algebraically closed field $\overline{k}$, (6.3.55)), applied to $K_{\overline{k}}/\overline{k}$, $t\otimes 1$, $f\otimes 1$. ■

6.3.11 Differentials are Weil Differentials, for General k

Fix a transcendental $t\in K$. By (6.3.32), every $D\in\mathrm{Der}_k(K,K)$ is $D(t)\cdot\partial_t$, so a differential $z\,\mathrm{d}t$ ($z\in K$) already carries all the information of an element of the (not yet formally constructed) module $\Omega_{K/k}$, and $\mathrm{Res}_P(z\,\mathrm{d}t)$ is available at every place (6.3.37). We show this recovers Ω_K exactly, generalising (6.3.31) beyond $k=\mathbb{C}$: the complex-analytic ingredients there (contour integration, Stokes' theorem, Mittag-Leffler) are replaced by the algebraic Residue Theorem, the trace formalism of (6.3.37), and the following fact, taken as given in this subsection.

Proposition 6.3.66 (Ω_K is a K-Vector Space)
Let K/k be an algebraic function field. For $x\in K$ and $\omega\in\Omega_K(D)$ ($D\in\mathrm{Div}(K)$), define $x\omega\in\mathrm{Hom}_k(\mathcal{A}_K,k)$ by $(x\omega)(\alpha):=\omega(x\alpha)$ (using that $\mathcal{A}_K$ is a K-algebra). Then $x\omega\in\Omega_K(D-(x))\subset\Omega_K$, and this action makes Ω_K a K-vector space.

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$x\omega \in \Omega_K(D-(x))$. For $\alpha \in \mathcal{A}_K(D-(x))$, $v_P(\alpha_P) \geq -v_P(D-(x)) = v_P(x) - v_P(D)$ for every P, so $v_P(x\alpha_P) \geq -v_P(D)$, that is $x\alpha \in \mathcal{A}_K(D)$; and $xK = K$. As $\omega \in \Omega_K(D)$ vanishes on $\mathcal{A}_K(D)+K$, it follows that $(x\omega)(\alpha) = \omega(x\alpha) = 0$ for every $\alpha \in \mathcal{A}_K(D-(x))+K$, that is $x\omega \in \Omega_K(D-(x))$.

Module axioms. For $x, y \in K$, $\lambda \in k$, $\alpha, \beta \in \mathcal{A}_K$: $(x\omega)(\alpha+\beta) = \omega(x\alpha + x\beta) = (x\omega)(\alpha) + (x\omega)(\beta)$ and $(x\omega)(\lambda\alpha) = \omega(\lambda x\alpha) = \lambda(x\omega)(\alpha)$, so $x\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ is again k-linear; $(x(y\omega))(\alpha) = (y\omega)(x\alpha) = \omega(yx\alpha) = ((xy)\omega)(\alpha)$ and $1\omega = \omega$; $((x+y)\omega)(\alpha) = \omega((x+y)\alpha) = \omega(x\alpha) + \omega(y\alpha) = (x\omega)(\alpha) + (y\omega)(\alpha)$; and $(\omega \mapsto x\omega)$ is k-linear by the same computation applied to $\omega_1 + \omega_2$. These are exactly the axioms making Ω_K a K-module, that is a K-vector space (being already a k-vector space). ∎

Proposition 6.3.67 (Ω_K is 1-Dimensional over K)
Let K/k be an algebraic function field. Then $\dim_K \Omega_K = 1$ (for the K-vector space structure of (6.3.66)).

Proposition 6.3.68 (Differentials are Weil Differentials)
Let K/k be an algebraic function field and $t \in K$ transcendental. For $z \in K$, define $\varphi_{z\,\mathrm{d}t} \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ by

$$\varphi_{z\,\mathrm{d}t}(\alpha) := \sum_P \mathrm{Res}_P(\alpha_P z\,\mathrm{d}t), \qquad \alpha = (\alpha_P)_P \in \mathcal{A}_K$$

(a finite sum: $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, and $z\,\mathrm{d}t$ is regular at all but finitely many P, so $\mathrm{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ for almost all P (6.3.45).c)). Then $\varphi_{z\,\mathrm{d}t} \in \Omega_K$, and

$$K \longrightarrow \Omega_K, \qquad z \longmapsto \varphi_{z\,\mathrm{d}t}$$

is a canonical K-linear isomorphism (for the K-vector space structures of (6.3.66) and K itself).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\varphi_{z\,\mathrm{d}t} \in \Omega_K$. Let D be the divisor with $v_P(D) := v_P(z\,\partial_\pi(t))$ for a uniformiser π at P (independent of the choice of π: for another uniformiser π', $\partial_{\pi'}(t) = \partial_{\pi'}(\pi)\,\partial_\pi(t)$ (6.3.34), with $\partial_{\pi'}(\pi)$ a unit at P, so $v_P(z\partial_{\pi'}(t)) = v_P(z\partial_\pi(t))$; and this vanishes for all but finitely many P, as z has finitely many poles and t finitely many poles). For $\alpha \in \mathcal{A}_K(D)$, $v_P(\alpha_P) \geq -v_P(D) = -v_P(z\partial_\pi(t))$ for every P, so $v_P(\alpha_P z\partial_\pi(t)) \geq 0$, that is $\alpha_P z\,\partial_\pi(t) \in \mathcal{O}_P$, giving $\mathrm{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ (6.3.45).c); so $\varphi_{z\,\mathrm{d}t}(\alpha) = 0$. For $f \in K$ embedded diagonally, $\varphi_{z\,\mathrm{d}t}(f, f, \ldots) = \sum_P \mathrm{Res}_P(fz\,\mathrm{d}t) = 0$ by (6.3.65) (the Residue Theorem), applied to $fz \in K$. So $\varphi_{z\,\mathrm{d}t}$ vanishes on $\mathcal{A}_K(D)+K$, that is $\varphi_{z\,\mathrm{d}t} \in \Omega_K(D) \subset \Omega_K$.

***K*-linearity.** k-linearity in z is immediate from (6.3.45).a), applied place by place and summed. For $x \in K$, using $x\alpha_P \cdot z = \alpha_P \cdot (xz)$ in the field K,

$$(x\,\varphi_{z\,\mathrm{d}t})(\alpha) = \varphi_{z\,\mathrm{d}t}(x\alpha) = \sum_P \mathrm{Res}_P((x\alpha_P)z\,\mathrm{d}t) = \sum_P \mathrm{Res}_P(\alpha_P(xz)\,\mathrm{d}t) = \varphi_{(xz)\,\mathrm{d}t}(\alpha),$$

so $x\,\varphi_{z\,\mathrm{d}t} = \varphi_{(xz)\,\mathrm{d}t}$: the map is K-linear (6.3.66).

Injectivity. Suppose $\varphi_{z\,\mathrm{d}t} = 0$ and $z \neq 0$; we derive a contradiction. Fix a place P and a uniformiser π at P. As ∂_π is a nonzero derivation and t is transcendental, $\partial_\pi(t) \neq 0$ (6.3.32), so $w := z\,\partial_\pi(t) \neq 0$ in K. Let $m := v_P(w)$ and $c := \pi^{-m-1} \in K^\star$; then $L_\pi(cw)$ has valuation exactly -1 (6.3.35), so

$$\gamma := \mathrm{Res}_P^{k(P)}(cz\,\mathrm{d}t) = L_\pi(cw)_{-1} \in k(P)$$

is its leading coefficient, and $\gamma \neq 0$. As k is perfect, $k(P)/k$ is separable (3.18.103), so by (3.18.131) there is $\bar{e} \in k(P)$ with $\mathrm{Tr}_{k(P)/k}(\bar{e}\gamma) \neq 0$; lift it to a unit $e \in \mathcal{O}_P^\star$ (any lift of a nonzero residue class is a unit, $\mathcal{O}_P$ being local). Then $L_\pi(e)$ has constant term $\bar{e}$ (6.3.35), and $L_\pi(cw)$ has no terms below X^{-1}, so the coefficient of X^{-1} in $L_\pi(ecw) = L_\pi(e)L_\pi(cw)$ is $\bar{e}\gamma$. That is $\mathrm{Res}_P^{k(P)}(ecz\,\mathrm{d}t) = \bar{e}\gamma$, and

$$\mathrm{Res}_P(ecz\,\mathrm{d}t) = \mathrm{Tr}_{k(P)/k}(\bar{e}\gamma) \neq 0.$$

Testing $\varphi_{z\,\mathrm{d}t}$ against the repartition supported at P with value $ec \in K$ then gives $\varphi_{z\,\mathrm{d}t}(\ldots) = \mathrm{Res}_P(ecz\,\mathrm{d}t) \neq 0$, contradicting $\varphi_{z\,\mathrm{d}t} = 0$.

Surjectivity. The map $K \to \Omega_K$ is K-linear and injective, and K is a 1-dimensional K-vector space over itself; so its image is a 1-dimensional K-subspace of Ω_K, which by (6.3.67) is itself 1-dimensional over K. So the image is all of Ω_K. ∎

© David Rufino CC BY-NC-ND 4.0

6.3.12 Tate's Approach to Residues: A Sketch

The route taken above defines Res_P through a uniformiser and the formal derivative, and proves the Residue Theorem (6.3.65) by reducing to $K = k(x)$ along a finite separable extension. Tate [Tat68] gives a different, intrinsic route, which we only sketch here (nothing later depends on it): the residue is defined from linear algebra alone, and the Residue Theorem is deduced from the finiteness results of the Riemann-Roch theory.

a) *Traces of finite-rank operators.* The trace of a finite-rank endomorphism T of a k-vector space V is its trace on any finite-dimensional subspace containing $T(V)$. It is additive, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ whenever $A : V \to W$, $B : W \to V$ and one of them has finite rank, and $\mathrm{Tr}(T) = \mathrm{Tr}(T|_H) + \mathrm{Tr}(\overline{T})$ if T preserves a finite-dimensional H, with $\overline{T}$ the induced operator on V/H. In particular $\mathrm{Tr}([T, F]) = 0$ for F of finite rank and T arbitrary.

b) *The local residue.* Let π be a k-linear retraction of the local space onto the lattice $\mathcal{O}_P$ (in Tate's setting the completion K_P, with π the projection killing the principal parts). For f in the local space let $T_f := \pi \circ (f \cdot -)|_{\mathcal{O}_P}$ be the Toeplitz operator on $\mathcal{O}_P$. Then $\big[T_f, T_g\big]$ has finite rank, because $(1-\pi)(f\mathcal{O}_P)$ lies in the finite-dimensional image of $t^{-n}\mathcal{O}_P/\mathcal{O}_P$, and $\mathrm{Res}_P(f\,\mathrm{d}g) := \mathrm{Tr}\big(\big[T_f, T_g\big]\big)$. Changing π changes each T_f by a finite-rank operator, which does not change the trace of the commutator. So no uniformiser, derivation or chain rule enters.

c) *Local properties.* It is k-linear, vanishes on differentials regular at P, has $\mathrm{Res}_P(\mathrm{d}t/t) = \deg P$ (the trace is taken over k) and is invariant under automorphisms of K_P; it should agree with the trace of the coefficient of t^{-1} used above.

d) *Global input.* The finiteness underlying Riemann-Roch: $H := K \cap \mathcal{A}_K(0) = L(0)$ and $\mathcal{A}_K/(K + \mathcal{A}_K(0))$ are finite dimensional (6.3.27); and $\mathcal{A}_K(D)/\mathcal{A}_K(0) = \bigoplus_P t_P^{-n_P}\mathcal{O}_P/\mathcal{O}_P$ is finite dimensional for every divisor $D = \sum_P n_P P$.

e) *The Residue Theorem as a trace identity.* Write $\mathcal{A} := \mathcal{A}_K$ and $L := \mathcal{A}(0) = \prod_P \mathcal{O}_P$, and for $f, g \in K$ let D be a divisor with $f\mathcal{A}(0), g\mathcal{A}(0) \subset \mathcal{A}(D)$.

   (a) For the retraction $\pi_L := \prod_P \pi_P : \mathcal{A} \to L$ and $T_f := \pi_L \circ (f \cdot -)|_L$, the commutator $\big[T_f, T_g\big]$ is the direct sum over P of the local commutators, of which only finitely many are nonzero (where f, g are regular the operators are multiplication by f, g and commute). So $\mathrm{Tr}\big(\big[T_f, T_g\big]\big) = \sum_P \mathrm{Res}_P(f\,\mathrm{d}g)$.

   (b) As in b), $\mathrm{Tr}\big(\big[T_f, T_g\big]\big)$ does not depend on the retraction $\pi : \mathcal{A} \to L$: another retraction changes T_f by $\psi \circ (1-\pi) f|_L$, which has finite rank since $(1-\pi)(\mathcal{A}(D))$ is finite dimensional.

   (c) Choose π adapted to K: write $K = H \oplus W_K$ and choose a finite-dimensional F with $\mathcal{A} = L \oplus W_K \oplus F$ (possible by d)), and let π be the projection onto L along $W_K \oplus F$. For $h \in H$, $fh \in K = H \oplus W_K$ so $T_f h \in H$: both T_f, T_g preserve H and act there as operators on a finite-dimensional space, whose commutator has trace 0.

   (d) On $L/H \cong (L+K)/K =: \Lambda \subset \mathcal{A}/K$ the induced operator $\overline{T}_f$ is the Toeplitz operator of multiplication by f on $\mathcal{A}/K$ (where f, g commute exactly), with respect to the retraction $\mathcal{A}/K = \Lambda \oplus \overline{F} \to \Lambda$. Writing $\overline{\pi} = 1 - p$ with p the projection onto $\overline{F}$, and using $fg = gf$ on $\mathcal{A}/K$, one finds $\big[\overline{T}_f, \overline{T}_g\big] = -\overline{\pi} f p g + \overline{\pi} g p f$ on Λ, whose trace equals $\mathrm{Tr}_{\overline{F}}([pg|_{\overline{F}}, pf|_{\overline{F}}]) = 0$, a commutator of operators on the finite-dimensional space $\overline{F}$.

   By (c), (d) and additivity of the trace, $\mathrm{Tr}\big(\big[T_f, T_g\big]\big) = 0$, so $\sum_P \mathrm{Res}_P(f\,\mathrm{d}g) = 0$ by (a) and (b).

Nothing here uses the topology of K_P. The construction of b) uses only the subspace $\mathcal{O}_P \subset K$, a retraction $\pi : K \to \mathcal{O}_P$, and $\dim_k t^{-n}\mathcal{O}_P/\mathcal{O}_P < \infty$ (as $k(\mathfrak{m}_P)/k$ is finite): "lattices" and "continuity" reduce to commensurability of subspaces and to the pole order of f, so the local residue may be defined on K itself, and the global step on the repartitions $\mathcal{A}_K$ of this book (components in K, not in the K_P) together with the finiteness of d). Compared with the proof above, this avoids the reduction to $k(x)$, the trace formula and separability, and works in any characteristic. What is left to check is that this Res_P agrees with the Laurent-coefficient residue used above (for a place of degree d, $\mathrm{Res}_P(\mathrm{d}t/t) = d$ with the trace taken over k); the global argument of e) is then complete.

6.3.13 Sheaf-Theoretic Interpretation of Repartitions

Unlike the rest of this chapter, this section depends on sheaf theory developed later in this book (4.2.2), and so is not logically self-contained at this point in the text; nothing later in this chapter depends on it.

Let $X := \mathbb{P}_K$, the set of places of K/k, equipped with the **cofinite topology** (opens are $\emptyset$ and complements of finite subsets); as $\mathbb{P}_K$ is infinite, any two nonempty opens intersect, so X is irreducible. Let $\mathcal{K}_X$ be the constant sheaf with value K (4.2.50). For $D \in \mathrm{Div}(K)$ define the subsheaf

$$\mathcal{O}(D)(U) := \{x \in K \mid v_P(x) \geq -v_P(D)\ \forall P \in U\} \subseteq \mathcal{K}_X(U)$$

generalising the structure sheaf ($\mathcal{O}(0) = \mathcal{O}_X$), with inclusion $\iota_D : \mathcal{O}(D) \hookrightarrow \mathcal{K}_X$.

© David Rufino CC BY-NC-ND 4.0

Proposition 6.3.69 (Stalk of $\mathcal{O}(D)$)
For $P \in X$, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is injective, identifying $\mathcal{O}(D)_P$ with the fractional ideal

$$\{y \in K \mid v_P(y) \geq -v_P(D)\} =: \pi_P^{-v_P(D)}\mathcal{O}_P$$

of K.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Stalk of $\mathcal{K}_X$. Since X is irreducible, $\mathcal{K}_X(U) = K$ for every nonempty open $U \subset X$, with every restriction map the identity (4.2.50). We can therefore represent the stalk as $\mathcal{K}_{X,P} = K$ with the identity maps. Under this identification, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is given by $(U, x) \mapsto x$.

Injectivity. Two representatives $(U, x), (U', x')$ of elements of $\mathcal{O}(D)_P$ become identified in the stalk exactly when they agree on some common refinement; as restriction maps are the identity, this means $x = x'$ already, as elements of K. So $(U, x) \mapsto x$ is injective, and we identify $\mathcal{O}(D)_P$ with its image in K.

$\mathcal{O}(D)_P \subseteq \{y : v_P(y) \geq -v_P(D)\}$. Every element of $\mathcal{O}(D)_P$ is represented by a pair (U, x), $P \in U$, $x \in \mathcal{O}(D)(U)$, whose image in $\mathcal{O}(D)_P$ is x itself (viewed in K); taking the defining condition of $\mathcal{O}(D)(U)$ at the point $P \in U$ gives $v_P(x) \geq -v_P(D)$.

$\{y : v_P(y) \geq -v_P(D)\} \subseteq \mathcal{O}(D)_P$. Let $y \in K$ with $v_P(y) \geq -v_P(D)$. By (6.3.9), y has finitely many poles, so together with finiteness of $\mathrm{Supp}(D)$, the set $T := \{Q \in X \mid v_Q(y) < -v_Q(D)\}$ is finite. As $P \notin T$, $U := X \setminus T$ is an open neighbourhood of P, and $y \in \mathcal{O}(D)(U)$ by construction of T; so (U, y) represents an element of $\mathcal{O}(D)_P$ mapping to y.

Fractional ideal. By the previous two paragraphs $\mathcal{O}(D)_P = \{y \in K \mid v_P(y) \geq -v_P(D)\}$. Writing π_P for a uniformiser of $\mathcal{O}_P$, (3.30.2) represents every $y \in K^\star$ uniquely as $y = u\pi_P^{v_P(y)}$ for $u \in \mathcal{O}_P^\star$, so $v_P(y) \geq -v_P(D) \iff y \in \pi_P^{-v_P(D)}\mathcal{O}_P$; thus $\mathcal{O}(D)_P = \pi_P^{-v_P(D)}\mathcal{O}_P$. This is the image of $\mathcal{O}_P$ under multiplication by the fixed element $\pi_P^{-v_P(D)} \in K$, hence an $\mathcal{O}_P$-submodule of K and so a fractional ideal of $\mathcal{O}_P$ (3.29.1) (indeed $\mathcal{O}_P$ is a discrete valuation ring by (6.3.7), so this is the fractional ideal $\mathfrak{m}_P^{-v_P(D)}$ of (3.30.7)); in particular it is a k-vector subspace of K, since $k \subset \mathcal{O}_P$ (6.3.5). ∎

Proposition 6.3.70 (Stalk of the Cokernel)
Since ι_D is a monomorphism (6.3.69), write $\mathcal{K}_X/\mathcal{O}(D) := \mathrm{coker}(\iota_D)$ (4.2.36). There is a unique morphism θ_P making

$$\begin{array}{ccc} & K & \\ \swarrow & & \searrow \\ (\mathcal{K}_X/\mathcal{O}(D))_P & \xrightarrow[\sim]{\theta_P} & K/\mathcal{O}(D)_P \end{array}$$

commute, and θ_P is (moreover) an isomorphism; in particular $(\mathcal{K}_X/\mathcal{O}(D))_P \cong K/\mathcal{O}(D)_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is (4.2.37), applied to $\iota_D : \mathcal{O}(D) \to \mathcal{K}_X$ (using $\mathcal{K}_{X,P} = K$), together with (6.3.69) identifying $\mathcal{O}(D)_P$ with $\pi_P^{-v_P(D)}\mathcal{O}_P$. ∎

Used freely below.

Proposition 6.3.71 (Repartitions compute the Cokernel Sheaf)
Writing $p_X : K = \Gamma(X, \mathcal{K}_X) \to \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ for the global sections of the cokernel map (4.2.35), there is an exact sequence of k-vector spaces

$$0 \longrightarrow L(D) \longrightarrow K \xrightarrow{p_X} \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \xrightarrow{\Phi} \mathcal{A}_K/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

where Φ sends $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ – a function $\tau : X \to \coprod_{P \in X} K/\mathcal{O}(D)_P$ (4.2.35) – to the class of any repartition $(\alpha_P)_P \in \mathcal{A}_K$ with $\alpha_P \equiv \tau(P) \pmod{\mathcal{O}(D)_P}$ for every P; that is, sending a representative $x \in K$ to its corresponding residue vector. In particular $i(D)$ is (via (6.3.71)) the dimension of the cokernel of p_X – via sheaf cohomology, the group $H^1(X, \mathcal{O}(D))$, a cohomological invariant of X, not merely of the algebra of repartitions.

Sketch. $(\iota_D)_U$ is the inclusion $\mathcal{O}(D)(U) \hookrightarrow K$, injective.

By (4.2.35) a global section of $\mathcal{K}_X/\mathcal{O}(D)$ is represented by a cover of X together with representatives in K over each open, and since the cofinite topology is quasi-compact (4.1.93) this cover may be taken finite, say $X = U_1 \cup \cdots \cup U_n$

© David Rufino CC BY-NC-ND 4.0

together with representatives $x_i \in K$, well-defined modulo $\mathcal{O}(D)(U_i)$, that **agree at every point of overlap**: for every place $P \in U_i \cap U_j$, the images of x_i, x_j in the stalk $(\mathcal{K}_X/\mathcal{O}(D))_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$ (6.3.70) coincide, that is

$$v_P(x_i - x_j) \geq -v_P(D)$$

Given such data, set $\alpha_P := x_i$ (a literal element of K, well-defined only up to the choice of i) for any i with $P \in U_i$. The class of α_P modulo $\pi_P^{-v_P(D)}\mathcal{O}_P$ is independent of the choice of i: if also $P \in U_j$, then $P \in U_i \cap U_j$, so the overlap condition above applies directly at this P and gives $v_P(x_i - x_j) \geq -v_P(D)$, that is $x_i \equiv x_j \pmod{\pi_P^{-v_P(D)}\mathcal{O}_P}$. Each x_i has finitely many poles (6.3.9), so $\{P \mid \alpha_P \notin \mathcal{O}_P\} \subseteq \bigcup_{i=1}^n \{P \in U_i \mid v_P(x_i) < 0\}$, a finite union of finite sets, is finite; so $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, giving $(\alpha_P)_P \in \mathcal{A}_K$.

Independence of the choice of representing data. Let $(U_i, x_i)_{i=1}^n$ and $(U_i', x_i')_{i=1}^m$ be two choices representing the same τ, with resulting repartitions $(\alpha_P)_P, (\alpha_P')_P \in \mathcal{A}_K$ as above. For each P, choosing i, i' with $P \in U_i \cap U_{i'}'$, both x_i and $x_{i'}'$ represent $\tau(P) \in K/\mathcal{O}(D)_P$ (4.2.35), that is $x_i \equiv x_{i'}' \pmod{\mathcal{O}(D)_P}$, so $\alpha_P - \alpha_P' = x_i - x_{i'}' \in \mathcal{O}(D)_P$. As this holds for every P, $(\alpha_P - \alpha_P')_P \in \mathcal{A}_K(D)$; so $(\alpha_P)_P$ and $(\alpha_P')_P$ have the same class in $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$. This assignment therefore descends to a well-defined map

$$\Phi : \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \longrightarrow \mathcal{A}_K/(\mathcal{A}_K(D) + K)$$

Exactness at K. For $c \in K$, $p_X(c)$ is represented by the trivial cover $\{X\}$ with $x_1 := c$, so $p_X(c) = 0$ if and only if $c \in \mathcal{O}(D)(X) = L(D)$; that is, $\ker(p_X) = L(D)$, matching exactness at K (the map $L(D) \hookrightarrow K$ being the inclusion).

Exactness at $\Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$. ($\mathrm{Im}(p_X) \subseteq \ker(\Phi)$) For $c \in K$, $p_X(c)$ has $\alpha_P = c$ for every P, that is, the constant repartition $\delta(c)$ (diagonal embedding $\delta : K \hookrightarrow \mathcal{A}_K$), whose class in $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$ is 0, since $\delta(c) \in K$; so $\Phi(p_X(c)) = 0$.

($\ker(\Phi) \subseteq \mathrm{Im}(p_X)$) Suppose $\Phi(\tau) = 0$, that is $(\alpha_P)_P \in \mathcal{A}_K(D) + K$, say $(\alpha_P)_P = \beta + \delta(c)$ with $\beta \in \mathcal{A}_K(D)$, $c \in K$. Then $x_i - c \in \mathcal{O}(D)(U_i)$ for every i (as $x_i - c = \alpha_P - c = \beta_P \in \mathcal{O}(D)_P$ for every $P \in U_i$), so τ and $p_X(c)$ are represented by the same local data, that is $\tau = p_X(c) \in \mathrm{Im}(p_X)$.

Exactness at $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$. Given $(\alpha_Q)_Q \in \mathcal{A}_K$, let

$$S := \{Q \mid \alpha_Q \notin \mathcal{O}_Q\} \cup \mathrm{Supp}(D) = \{P_1, \ldots, P_n\}$$

finite, being a union of two finite sets; note $v_Q(D) = 0$ for every $Q \notin S$, so $\mathcal{O}(D)_Q = \mathcal{O}_Q$ there. Set $x_0 := 0$ and, for $i = 1, \ldots, n$, $x_i := \alpha_{P_i}$ and $Z_i := \{Q \neq P_i \mid \alpha_{P_i} \notin \mathcal{O}_Q\}$, the poles of α_{P_i} other than possibly at P_i itself, finite by (6.3.9). Since the cofinite topology has no "small" neighbourhoods – any nonempty open already excludes only finitely many points – each U_i ($i \geq 1$) must be taken to also avoid x_i's own poles, not merely to contain P_i: set

$$U_0 := X \setminus S, \qquad U_i := X \setminus ((S \cup Z_i) \setminus \{P_i\}) \quad (i = 1, \ldots, n)$$

These cover X: $U_0 \cup \cdots \cup U_n \supseteq (X \setminus S) \cup \{P_1, \ldots, P_n\} = X$, as $P_i \in U_i$ for each i.

Each U_i meets S in at most the single point P_i: $U_0 \cap S = \emptyset$, and $U_i \cap S = \{P_i\}$ for $i \geq 1$, since U_i excludes every other point of S. Consequently, for $i \neq j$, no point of S lies in both U_i and U_j (it would have to be the marked point of each, and the marked points – or absence thereof, for $i = 0$ – differ), so $U_i \cap U_j \subseteq X \setminus S$; in particular $Q \in U_i \cap U_j$ gives $v_Q(D) = 0$ and $Q \neq P_i, P_j$.

They agree at overlaps: for $i \neq j$ in $\{0, \ldots, n\}$ and $Q \in U_i \cap U_j$, $v_Q(x_i) \geq 0$ (trivially if $i = 0$, as $x_0 = 0$; else since $Q \in U_i$, $Q \neq P_i$ gives $Q \notin Z_i$) and likewise $v_Q(x_j) \geq 0$; so by the ultrametric inequality $v_Q(x_i - x_j) \geq \min(v_Q(x_i), v_Q(x_j)) \geq 0 = -v_Q(D)$.

This data therefore glues to a class $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ with $\tau(Q) = \alpha_Q \pmod{\mathcal{O}(D)_Q}$ for every Q (on U_0 this matches $x_0 = 0$, since $Q \notin S$ gives $\alpha_Q \in \mathcal{O}_Q = \mathcal{O}(D)_Q$; at $Q = P_i$ this matches $x_i = \alpha_{P_i}$ directly), so $\Phi(\tau) = (\alpha_Q)_Q$ exactly; so Φ is surjective, matching exactness at the final term. ∎

© David Rufino CC BY-NC-ND 4.0